Scoring Library Classification Schemes with Hierarchical-Clustering Objectives: A Survey of Open Schemes and a Common Benchmark

David Awad

Working draft. DRAFT, not submitted.

Abstract

Library classification schemes are compared mostly by description: their history, notation, coverage and licence. Quantitative comparisons exist, but they measure retrieval effectiveness, the scatter of a subject across schemes or coverage of a reference map of knowledge. We found no study that scores the hierarchy of several universal or near-universal schemes on one structural objective over the same set of works. We survey thirteen open or openly obtainable scheme builds and propose such a comparison, built on objectives from the theory of hierarchical clustering, and we test how far it can be trusted. A scheme, restricted to the classes a corpus uses, is a rooted tree over the works. We score that tree with Dasgupta's cost and Moseley and Wang's revenue on a fixed k-nearest-neighbour graph of text embeddings, and normalize revenue against an average-linkage tree on the same graph; the ratio is called skill. The corpus is 10,000 arXiv papers with CC0 metadata, a holdout of 500 Open Library editions with librarian-assigned LC call numbers and a sample of 3,000 books that also carry Dewey numbers. We keep source-assigned ("gold") placements apart from placements made by one lexical classifier. On gold labels, arXiv's own taxonomy reaches skill 0.786 on arXiv papers and LCC reaches 0.419 on the books. With the classifier's placements the same two schemes fall to 0.128 and 0.005. The findings are mostly cautions. Coverage drives most classifier-placed rankings: across 13 schemes skill correlates with coverage at Spearman $\rho = 0.76$, and coverage explains 59% of the rank variance of cost. The metric agrees only weakly with a blind LLM judge ($\kappa = 0.27$ on 953 pairs); a false-join term raises κ to 0.48, but not agreement with citation or subject-heading signals. Rankings are stable across embedding model, k and graph threshold within one classifier (minimum ρ 0.90) and not across classifier tiers (minimum 0.04), and objectives disagree about FDC against LCC. On 3,000 books, Dewey sections against LCC is a tie that depends on the setting. Under a coverage-floor rule, the LCC outline with LCSH vocabulary is the default among open schemes in 27 of 27 book settings, partly through its vocabulary. A fitted order-preserving scheme leads every human scheme on its own graph; the lead survives rescoring on two independent embedding models, part of it is home advantage, and on books it does not survive the false-join correction. We also report a classifier defect that the benchmark exposed.

Keywords: library classification; hierarchical clustering; Dasgupta cost; knowledge organization; open data; benchmark

1 INTRODUCTION

A universal classification scheme gives every work one address in a single tree. The Library of Congress Classification (LCC), the Dewey Decimal Classification (DDC), the Universal Decimal Classification (UDC), the Basic Classification (BK), the Integrative Levels Classification (ILC) and others each claim to arrange all of knowledge. A practitioner choosing among them, or a researcher designing a new one, needs to ask which arrangement keeps related works close together, and at what navigation cost. The literature answers this mostly qualitatively, by notation, facet structure, history and institutional adoption [1], [2], [3], [4]. Quantitative comparisons have measured retrieval performance [5], [6], [7] or the scatter of a subject across schemes [8], [9], not the structure of the hierarchy itself (Section 1.1). A structured search of JASIST, Knowledge Organization, Cataloging & Classification Quarterly, ISKO and SIG/CR proceedings, JCDL and TPDL (project report 8, `research/reports/8-prior-work-search.md`) found no study that scores several universal schemes on a common hierarchical objective over the same corpus, and no use of Dasgupta cost or Moseley–Wang revenue on a human-built scheme. The search covered titles and abstracts, not full texts (Section 12), so this is absence of evidence, not proof.

Three developments make the question more pressing.

1. **The main lookup service is gone.** OCLC Classify returned the most common DDC and LCC numbers for a work from WorldCat holdings. OCLC discontinued it on 31 January 2024 [10]. No free service replaced it.
2. **Automatic placement has not closed the gap.** Annif, the most mature open subject-indexing toolkit, is trained separately for each vocabulary [11]. On more than 230,000 Swedish union-catalogue records its best (ensemble) configuration reaches 66.82% accuracy at three-digit DDC [12]. Recent work that uses large language models (LLMs) for LCC and DDC assignment reports weakness with more granular classes [13] and a lack of specificity [14], and prompted LLMs fall well short of fine-tuned smaller models on DDC classes for research data [15]. The Library of Congress's own experiments predicted subject and genre with under 50% accuracy, short of a

95% automation threshold [16]. Placement is getting cheaper, so the open question shifts towards *which scheme* to place into.

3. **Licences constrain open use.** DDC is copyrighted by OCLC [17]. The full UDC is licensed [18]. Openly licensed universal schemes are shallow, such as BK [19] and the Free Decimal Correspondence (FDC) [20], or their licence terms are unsettled, as for ILC2 [21]. Data-driven vocabularies such as OpenAlex Topics are open but cover research literature only [22]. A quantitative comparison helps decide whether an open scheme is good enough to adopt.

Contributions. (i) A survey of the thirteen scheme builds benchmarked here, with DDC and UDC as context, covering structure, size, maintainer and redistribution status (Section 2). (ii) A scheme-agnostic scoring method. It induces a tree from a scheme and a set of placements, and scores the tree with Dasgupta cost and Moseley–Wang revenue on a fixed similarity graph. The score is normalized against an average-linkage reference tree on the same graph, and coverage, structure and navigation statistics are reported with it (Section 3). (iii) A pinned, CC0-metadata corpus of 10,000 arXiv papers and a 500-edition LC holdout (Section 4). (iv) Results for 13 scheme builds. Gold placements and classifier placements are kept apart, and the classifier bias is calibrated where both exist (Section 5). (v) A methods note on a classifier defect that the benchmark exposed, with its before-and-after effect (Section 6). (vi) A short account of an order-preserving scheme fitted to the same objective, with its caveats (Section 10). (vii) A validity check of the cost against a blind judge, a false-join correction and non-LLM checks (Section 7); a coverage control, a second classifier tier and a rank-stability sweep (Section 8); and a book comparison of Dewey and LCC that is a setting-dependent tie (Section 9).

1.1 Related work

Comparing classification schemes. The Cranfield project compared the Universal Decimal Classification, an alphabetical subject catalogue, a faceted classification and Uniterm coordinate indexing, applied to 18,000 aeronautics documents [5], [23]. Keen compared five index languages built for a test collection on retrieval effectiveness [6], and Carpenter et al. compared five patent classifications on recall and precision for the same questions [7]. O'Neill et al. measured how the classes of LCC and DDC disperse into each other on MARC records [8], and Connaway and Sievert compared the scatter of one subject across LCC, DDC and the NLM Classification [9]. Zins and Santos mapped the top two levels of LCC, DDC and UDC onto one reference model of knowledge [2]. In bibliometrics, research classifications are scored on citation-based criteria: algorithmic journal maps against the ISI Subject Categories [24], the Web of Science and Scopus journal classifications against each other [25], and taxonomies built from different citation relations against one gold standard [26]. These studies score schemes by retrieval, by the scatter of a mapping or by citation coherence. None scores the tree itself. For a single vocabulary, corpus-weighted pruning measures how much of a subject tree a collection needs [27], checking functions detect quality defects in SKOS vocabularies [28], and structural analysis is one of the established ways to evaluate a thesaurus [29].

Objectives for hierarchical clustering. Dasgupta defined the cost of a tree T over items with pairwise similarities w_{ij} as $\sum_{i<j} w_{ij}\,|\mathrm{leaves}(T[i \vee j])|$, where $i \vee j$ is the lowest common ancestor (LCA) of i and j [30]. Similar pairs should be separated as deep in the tree as possible. A flat tree has the maximum cost. Moseley and Wang studied the dual, revenue, $\sum w_{ij}(n - |\mathrm{leaves}(T[i \vee j])|)$, so that cost plus revenue equals $n \sum w_{ij}$. They showed that average linkage is a $\frac{1}{3}$-approximation to the maximum revenue [31], [32]. Optimizing Dasgupta cost exactly is NP-hard. Approximation algorithms use sparsest cut and spreading metrics [33], [34]. Cohen-Addad et al. give an axiomatic family of admissible objectives and show when average linkage and its variants recover planted hierarchies [35]. Hyperbolic embeddings give a differentiable relaxation [36], and structural constraints can be added to the search [37]. scikit-network ships a Dasgupta cost and score for binary dendrograms [38]. This line of work treats the objective as something to *optimize*. We use it first to *evaluate* trees that people built.

No objective-free optimum. Kleinberg's impossibility theorem says no clustering function satisfies scale invariance, richness and consistency together [39]. Carlsson and Mémoli characterize single linkage axiomatically among hierarchical methods [40]. One consequence for us: a score is always relative to the similarity function. We therefore fix and publish the graph and treat it as part of the measurement.

Navigation. Optimal search trees and Huffman coding minimize expected depth under an access distribution [41], [42]. The hierarchical-identity model of Watts, Dodds and Newman makes greedy navigation on a hierarchy a measurable success criterion [43]. Basic-level categories explain why a natural middle level carries most human use [44]. User studies of menu hierarchies disagree on the best shape. Speed and accuracy improved with breadth in one study [45], and task time rose with depth for web links [46], but others found an optimum at moderate breadth [47], [48]. Landauer and Nachbar fit the time per menu screen as $k + c \log b$ for b choices [49], the form of our $\mathrm{nav}_{\log}$ model. For a two-level card file, the optimum number of records per guide is proportional to the square root of the file size [50]. We report simple, uniform-use navigation costs and fit no user model.

Data-driven and LLM-built taxonomies. Waltman and van Eck cluster publications on direct citation at several resolutions [51], using the Leiden algorithm [52]. OpenAlex Topics come from citation clustering and are named by an LLM, and their upper levels are mapped from Scopus ASJC categories [22], [53]. TaxoGen and TaxoExpan build and extend topic taxonomies from text

Table 1. The classification systems: thirteen benchmarked builds, with DDC and UDC as context. "Classes" is the number of classes in the build used in the benchmark. "Bundled" means the data file is committed in the repository.

scheme	maintainer	version	classes	structure	licence and redistribution status
`lcc`	Library of Congress	2024	7,392	21 main classes (letters); subclasses and numbered spans (outline)	US government work, 17 U.S.C. 105; LC's Classification Web notice calls LC classification copyrighted outside the USA. Bundled
`lcc-full`	Library of Congress	2024	416,186	13 levels; 408,794 numbered entries from 44 FreeLCC text PDFs added to the outline; max 7,994 children	As `lcc`. Bundled
`fdc`	J. M. Ockerbloom	0.06 (2010-08-19)	1,097	6 levels; Dewey-shaped numbers 000 to 999.9999; written from DDC 11th ed. (1922)	CC0 1.0. Frozen; author states no long-term maintenance. Bundled
`bk`	GBV (K10plus)	export 2026-10-03	2,140	Five roots, two-digit headings, decimal classes; academic bias	CC0 1.0. Bundled; weekly rolling export
`ilc2`	ISKO Italy (C. Gnoli and contributors)	2019-10-17	10,117	Faceted, ordered by integrative levels; 9,174 concepts and 943 facet indicators	Unsettled: CC BY 3.0 in the SKOS file, CC BY-SA 3.0 in BARTOC, no site page states a licence. Not bundled
`arxiv`	arXiv (arxiv-base)	2024-04-01	192	8 groups, 38 archives (18 retired), categories	MIT (arxiv-base). Bundled
`msc`	Mathematical Reviews (AMS) and zbMATH Open	2020	6,603	63 sections, subsections, leaf codes	CC BY-NC-SA 4.0. Not bundled (built from the official CSV)
`jel`	American Economic Association	2026	1,015	20 categories, subcategories, codes (A, A1, A10)	All rights reserved; terms grant personal use only. Not bundled
`acm-ccs`	ACM	2012	2,113	13 top-level concepts, depth up to 6; 159 concepts have more than one parent	ACM copyright; "freely available for educational and research purposes". No redistribution grant. Not bundled
`ipc`	WIPO	2026.01	80,145	8 sections, 132 classes, 655 subclasses, 7,668 main groups, 71,682 subgroups	CC BY 4.0 (WIPO Open Access Policy). File too large to commit; built by a pinned builder
`openalex-topics`	OpenAlex	2026-10-09	4,798	4 domains, 26 fields, 252 subfields, 4,516 topics; citation-derived, LLM-named	CC0 1.0. Bundled
`unesco-thesaurus`	UNESCO	2026-08-31	4,595	7 domains, 88 microthesauri, about 4,500 concepts; 229 concepts have 2 to 6 broader terms	CC BY-SA 3.0 IGO. Adaptation stays share-alike
`eurovoc`	Publications Office of the EU	4.24 (2026-07-08)	7,663	21 domains, 127 microthesauri, 7,515 descriptors; 8 levels	Sources disagree (Commission reuse notice, CC BY 4.0 for editorial content, CC0 in BARTOC). Not bundled
`DDC`	OCLC	ed. 23	n/a	Decimal tree; 1,000 classes at three digits	Copyright and trademark OCLC; full schedules only through paid WebDewey
`UDC`	UDC Consortium	Summary	n/a	Faceted decimal; Summary has about 2,600 of 70,000+ entries	Full reference file licensed to restricted communities; Summary CC BY-NC 4.0 from 2026 (see text)

[54], [55]. LLM pipelines generate layers or labels: Chain-of-Layer, TnT-LLM and TaxoAdapt [56], [57], [58]. None of these optimizes a global objective. Taxonomy-quality work catalogues attributes and measures [59]. It includes a reference-free robustness score that correlates semantic and taxonomic similarity [60], which is closest in spirit to ours.

Automatic subject classification. Annif [11], its SemEval-2025 Task 5 entry [61] and the task overview [62] cover subject indexing with thesauri such as GND. They report results per vocabulary, not across schemes. An open LCC classifier covers 15 of the 21 top-level classes [63]. We found no public leaderboard that compares classification schemes, as opposed to classifiers.

2 A SURVEY OF THE CLASSIFICATION SYSTEMS

This section describes the thirteen scheme builds that the benchmark scores, and gives DDC and UDC as context. The benchmark uses twelve distinct schemes: LCC enters twice, as the published outline (`lcc`) and as the complete schedules (`lcc-full`). Facts about structure and size are those recorded by the scheme builders in the repository. Facts about licences are those recorded in the scheme READMEs and in the licensing report of the project (checked 9–10 October 2026); they are a description of what each publisher states, not legal advice.

2.1 Library classifications: LCC, DDC, UDC

LCC is maintained by the Library of Congress. Its schedules are published as PDFs for letters A–Z [64], and the Library describes its linked-data service as a public-domain data set [65]. The outline has 21 main classes (the letters A–Z less I, O, W, X and Y). The complete schedules are not available as one machine-readable current file: the project's report found a 2016 MARC bulk snapshot of 649,332 records, about 2.5 million per-URI records on id.loc.gov, and the PDFs, which the `lcc-full` builder parses. Whether LCC is free to redistribute depends on jurisdiction. US government works are not subject to copyright [66], but LC's Classification Web notice says LC-originated classification data are copyrighted except within the USA [67]. LCC has been developed in successive editions since the early twentieth century [68].

DDC is owned by OCLC [17]. Full schedules are available through the subscription WebDewey, and OCLC's terms for Dewey linked data prohibit scraping, indexing and long-term storage; the base set is free to view only. The DDC 22 Summaries PDF carries "All rights reserved". DDC schedules and captions are therefore absent from the

benchmark, although OCLC describes DDC as the world's most widely used way to organize library collections [17]. The later book comparison (Section 9) uses only Dewey class *numbers* taken from public catalogue records, with the tree built from the digit prefixes of the notation and no caption, schedule or table. That is an engineering position, not legal advice, and it is written for the United States and the European Union only. The FDC was written from the out-of-copyright 1922 eleventh edition of DDC and checked only for compatibility.

UDC is maintained by the UDC Consortium. The full Master Reference File is licensed, with several licence types for restricted communities, and embedding it in software requires a paid publishing licence [18]. The licensing report records the UDC Summary (about 2,600 of 70,000+ entries) as CC BY-NC 4.0 from 2026, where earlier search snippets and the survey report described a share-alike Creative Commons 3.0 licence. We use the later statement. Either way the non-commercial condition, or the share-alike condition, blocks use inside a permissively licensed tool.

2.2 Open universal and near-universal schemes

No open-licensed scheme is at once universal, deep and equipped with a full LCC or DDC crosswalk. The open candidates are shallow, or are not shelf orders. The survey report records the main ones as follows.

BK (Basisklassifikation) is a hierarchical decimal subject classification for scholarly works, maintained by the Gemeinsamer Bibliotheksverbund [19]. It has 2,140 classes in this export (BARTOC lists 2,139, and the export carries one class marked as deleted). Its data are CC0 and refreshed weekly. It is academically biased, and its crosswalk is thin: the survey counted 130 mappings from BK in coli-conc, of which 49 point to DDC edition 23 and none to LCC. **FDC** is CC0 and DDC-shaped, but frozen since 2010 [20]. **ILC2** [21] is universal and faceted, orders phenomena by integrative levels rather than by library convention, and has no settled licence. Its SKOS file declares CC BY 3.0 while the BARTOC record lists CC BY-SA 3.0, and no page on the maintainers' site states a licence; the project therefore treats the data as CC BY-SA 3.0 if redistributed.

OpenAlex Topics is a four-level, citation-derived and LLM-named hierarchy for research literature only, released as CC0 [22], [53], [69]. **UNESCO Thesaurus** and **EuroVoc** are thesauri, narrower than a universal scheme, with a domain–microthesaurus–descriptor layout. **IPC** is the International Patent Classification, a deep scheme that is relevant here as a large hierarchy over a technical domain. WIPO's open access policy issues its online content under CC BY 4.0 unless marked otherwise, and the master files carry no other notice [70], [71]. The remaining schemes are domain schemes: arXiv's category taxonomy [72], MSC2020 [73], JEL [74] and ACM CCS [75]. Of these, only arXiv's carries an open software licence, and JEL publishes no open licence.

ACM CCS, 1998 and 2012. The 2012 system replaced the 1998 edition. ACM says it mapped the old scheme to the new one and shows both terms on article citation pages, but publishes no 1998-to-2012 mapping file; its public downloads (SKOS, Word and HTML) contain the 2012 tree only, and the SKOS file carries no match relations, notes or 1998 codes. The only public trace of the mapping is the pair of terms on each article's page, which ACM's policy does not allow scripts to harvest. We found no mapping file and did not build one. This matters below, because the arXiv `acm-class` field of the corpus carries CCS 1998 codes (Section 4).

Other surveyed resources. The survey also recorded resources that are supplements rather than shelf orders: Wikidata (CC0, a class graph, not a shelf order), FAST (ODC-By headings), LCSH (headings with no class numbers), GND and Thema. Wikidata is CC0 and holds 942 items with both a DDC and an LCC statement, the only open DDC–LCC concordance the survey measured, and a small one. The Open Shelves Classification was never completed. The survey found no open machine-readable copy of the Colon, Bliss or ICC classifications.

3 METHOD

3.1 Similarity graph

Each item's text is the title and abstract joined by a full stop (titles only for the book holdout, which has no descriptions). It is embedded with `sentence-transformers/all-MiniLM-L6-v2` at a pinned ONNX revision [76], [77], [78], [79], and vectors are L2-normalized. Raw cosine similarities between unrelated texts are not centred on zero, so we subtract the exact mean cosine τ over all unordered pairs: $\tau = 0.109$ on the arXiv corpus and $\tau = 0.173$ on the holdout. Each item keeps its $k = 20$ most similar items. The weight of an edge is $w_{ij} = \max(0, \cos_{ij} - \tau)$. The graph is symmetrized by taking the maximum, and zero-weight edges are dropped. That leaves 148,111 edges on the corpus and 7,297 on the holdout. Every scheme and the reference tree are scored on this one graph, so all comparisons on a corpus are paired. The classifier's optional embedding tier uses a different model (BGE-small, not used in this run). This follows the benchmark design, which aims to avoid a circular advantage for schemes whose labels embed well.

3.2 From a scheme and placements to a tree

A scheme is a set of classes, each with at most one parent, so it is a tree or a forest. For polyhierarchies such as EuroVoc and the UNESCO Thesaurus, the port has already chosen one primary parent per concept. Given one placement per item, we keep the used classes and all their ancestors. Forest roots hang under a virtual root. Each item becomes a leaf under its class node, including when that class is an internal node. An item with no placement becomes a leaf directly under the root, so every pair involving it pays the full corpus size. Coverage is therefore penalized *inside* the cost, and we also report it separately. All reported runs use one class per item: the first listed code for gold labels and the classifier's top choice other-

wise. The harness also supports fractional (multi-copy) placement, which we leave for later work.

3.3 Cost, revenue and skill

Let n be the number of items (placed or not), E the edges of the graph, $W = \sum_{(i,j)\in E} w_{ij}$, and $N(u)$ the number of items in the subtree of node u. Then

$$\begin{aligned} \text{cost}(T) &= \sum_{(i,j)\in E} w_{ij} N(\text{lca}_T(i,j)), \\ \text{cost}_{\text{norm}} &= \frac{\text{cost}(T)}{nW}, \\ \text{revenue}_{\text{norm}} &= 1 - \text{cost}_{\text{norm}}. \end{aligned} \tag{1}$$

A tree with every item under the root has $\text{cost}_{\text{norm}} = 1$. By the Moseley–Wang identity, $nW - \text{cost}(T)$ is the revenue, so $\text{revenue}_{\text{norm}}$ is revenue over its trivial upper bound. The sum runs over graph edges only. This is a kNN-sparsified Dasgupta cost: the pairs that matter are those the embedding calls close.

Revenue depends on the corpus and the graph, so we normalize it against a reference tree T_{ref}. That tree is a deterministic average-linkage (UPGMA) dendrogram on the same graph, with missing edges treated as similarity 0:

$$\text{skill}(T) = \frac{1 - \text{cost}_{\text{norm}}(T)}{1 - \text{cost}_{\text{norm}}(T_{\text{ref}})}. \tag{2}$$

Skill is 0 for the flat tree and 1 for a tree as good as average linkage. Average linkage is a $\frac{1}{3}$-approximation of maximum revenue [32], so the reference is an achievable tree, not an optimum. Skill may exceed 1. The reference reaches $\text{cost}_{\text{norm}} = 0.152$ on the corpus and 0.301 on the holdout.

The reference is a binary tree over individual items. A scheme's tree stops at classes, and all items inside one class share an LCA. This tie is intended: it models scanning the class shelf. It also means that no scheme whose classes hold more than one item can reach skill 1 easily. Skill compares a scheme with a data-fitted tree. It is not a pass mark.

3.4 Gold and shelfmark-placed subsets

A score depends on the scheme *and* on how works were placed into it. We keep two subsets apart.

- **Gold (track A).** Labels supplied by the source: arXiv primary categories, author-supplied MSC and JEL codes on the arXiv corpus, and LC call numbers on the book holdout. Call numbers are resolved to the deepest covering class of the bundled LCC outline (`lcc`) and of the complete LCC schedules (`lcc-full`). The resolver tries the full call number, then the class number without cutters, then the class letters.
- **Placed (track P).** Labels assigned by shelfmark's lexical classifier (`profile:lexical@2`). Each scheme gets an isolated store with no hand-written rules. Placements below the classifier's 0.8 confidence target are kept and flagged. Fallback placements, where no tier produced a candidate, count as unplaced.

The two subsets are never ranked against each other. Where both exist for the same scheme and items (arXiv, MSC and JEL on the corpus; LCC and `lcc-full` on the holdout), their difference measures the classifier's effect (Section 5.3).

3.5 Structure, coverage and navigation

For each run we report:

- **Coverage:** the share of items with a placement.
- **Depth:** the mean, median, 95th percentile and maximum item depth, in edges from the root to the item leaf. The root is depth 0, and an item placed in a top-level class has depth 2.
- **Branching:** the fan-out of used internal nodes. Item leaves count as children, so unplaced items inflate root fan-out.
- **Balance:** the effective number of classes $\exp(H)$ and the Gini coefficient. Both use the distribution $p_u \propto N(u)$ over used class nodes at *all* levels, so an item counts once at each ancestor class. These are multi-level summaries, not leaf-partition statistics.
- **Expected navigation cost** under uniform use, with two menu models. For an item i with ancestors u of fan-out $b(u) > 1$,

$$\text{nav}_{\log}(a) = \frac{1}{n} \sum_i \sum_u (a + \log_2 b(u)) \tag{3}$$

  is a Hick–Hyman-style decision cost with per-level overhead $a \in \{0, 1, 2\}$. The second model is

$$\text{nav}_{\text{scan}} = \frac{1}{n} \sum_i \left[\sum_u \frac{b(u)+1}{2} + \frac{n_c + 1}{2} \right], \tag{4}$$

  a linear menu scan plus a scan of the n_c items under the item's deepest class. These parameters are assumptions, not fitted values. We report the sweep over a.

3.6 Implementation

The harness is a Rust crate, `shelfmark-bench`. It walks parent pointers to find LCAs, which the shallow class trees make cheap, and computes the average-linkage reference once per graph with a size-weighted (Lance–Williams) update and deterministic tie-breaking. Unit tests check a hand-computed cost, the identity cost + revenue $= nW$, that a root-only tree has $\text{cost}_{\text{norm}} = 1$, and that the embedder, the graph and the reference are deterministic. Each result records the SHA-256 of the graph, the induced tree and the scheme file.

4 CORPUS

arXiv corpus. The arXiv OAI-PMH endpoint (`arXivRaw`) was harvested for records with datestamps from 2026-01-01 to 2026-10-09 in eight top-level groups. A seeded SHA-256 ranking then sampled 10,000 records, proportional to group size with a floor of 100 per group: cs 3,968 (39.7%), physics 2,630 (26.3%), math 2,042 (20.4%), eess 519 (5.2%), stat 345 (3.5%), econ 185 (1.9%), q-bio 172 (1.7%) and q-fin 139 (1.4%). arXiv applies CC0 1.0 to metadata [80]. Datestamps follow record updates, so older papers appear: 69.7% of identifiers carry a 2026 prefix. Computer science, mathematics and physics make up 86.4% of the sample. Gold labels in other schemes are

Table 2. Gold placements on the arXiv corpus. The `msc` row is the corrected gold run (see text); `arxiv` and `jel` are from the original run.

scheme	cov.	cost_norm	revenue _norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log ($a = 1$)
`arxiv`	100.0%	0.333	0.667	0.786	36.5	3.90	4	797	0.795	18.3
`msc`	13.2%	0.984	0.016	0.018	461.5	1.39	4	25	0.599	15.2
`jel`	0.02%	1.000	0.000	0.000	6.0	1.00	4	1	0.000	14.3

Table 3. Gold placements on the book holdout. LC call numbers are resolved to the deepest covering class.

scheme	cov.	cost_norm	revenue _norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log ($a = 1$)
`lcc-full`	92.2%	0.706	0.294	0.421	85.2	5.19	10	68	0.718	19.7
`lcc`	92.2%	0.708	0.292	0.419	35.0	4.20	9	89	0.791	18.6

sparse: 1,325 papers carry an MSC 2020-shaped code, 317 an ACM CCS 1998 code and 2 a JEL code (after the correction of the label fields described in Section 5, which leaves the text, order and identifiers unchanged).

Open Library holdout. 500 Open Library edition records with a non-empty `lc_classifications` array were drawn through the Open Library Search and editions APIs [81]. They hold titles only, with no description: 0 rows have text beyond the title, and titles average 3.5 words. The holdout is skewed: 382 records (76.4%) have a first call number in class P (language and literature). The other 14 letters seen are B 32, H 19, Q 8, D 5, M 4, R 3, U 2, N 2, C 2, J 1, G 1, E 1, L 1. 37 records hold only an empty string in the array. Open Library asks contributors to share under CC0 1.0 [82]. Both files are pinned by SHA-256 and by ID lists.

Book sample with Dewey numbers. A second book file holds 3,000 editions that carry an LC call number, a Dewey number and subjects, with equal quotas over the 21 LCC main classes (Section 9). It shares no work with the 500-book holdout.

Schemes. Table 1 lists the 13 builds with their class counts and licence status. Five are general or universal in intent: LCC (outline and complete schedules), FDC, BK and ILC2. The rest are thesauri or vocabularies (UNESCO Thesaurus, EuroVoc, OpenAlex Topics) or domain schemes (arXiv, MSC, JEL, ACM CCS, IPC). Each scheme file's SHA-256 is recorded, and the rerun under the corrected classifier reused byte-identical files.

5 RESULTS

Every table in this section is generated from the committed result files (Section 13). Placed rows are sorted by $\text{cost}_{\text{norm}}$, best first. All scores are on the graphs of Section 3.1; skill is relative to the average-linkage reference.

5.1 Gold placements

arXiv's own taxonomy, applied to arXiv papers by their authors, reaches skill 0.786 with full coverage (Table 2). This is the highest score in the study. We read it as an upper reference for what a well-fitted scheme with correct placements achieves on this graph, not as evidence that arXiv's taxonomy is a good *universal* scheme. The arXiv categories and the abstracts come from the same authors, and the corpus was sampled by arXiv group.

On the books, LCC call numbers resolved into the outline reach skill 0.419 and into the complete schedules 0.421, at the same coverage (92.2%, Table 3). The complete schedules use more effective classes (85 against 35) and are deeper (mean depth 5.19 against 4.20). Yet cost barely moves (0.706 against 0.708). With 500 short titles and a holdout that is three quarters class P, the title-embedding graph does not reward the finer LCC distinctions. This does not mean the distinctions are useless; it is a limit of what the graph can see.

MSC and JEL gold results are uninformative. With 13.2% and 0.02% coverage, almost every pair involves an item under the root. The MSC gold run was rescored after a correction of the corpus's `msc` field (it had held markers and words from the `msc-class` text) and of the MSC normalizer, which now accepts section and form codes such as `65-02`. Coverage moved from 13.1% to 13.2% and skill from 0.0183 to 0.0185. The corrected run is reported here. ACM CCS still has no gold run. The corpus's `acm_ccs` field originally held the MSC-style `msc-class` text copied from arXiv's ACM-class field. After the correction it holds ACM CCS 1998 codes such as `I.2.6` on 317 papers, but the scheme is CCS 2012, whose codes are paths of 8-digit concept IDs, and no 1998-to-2012 crosswalk is available (Section 2). The run log records the run as skipped for that reason.

5.2 Placement by the lexical classifier

On the corpus, skill under classifier placement runs from 0.059 (msc) to 0.423 (openalex-topics). Second is unesco-thesaurus at 0.329. Coverage explains much of the order: across the 13 schemes the Spearman correlation between coverage and skill is 0.76 (Figure 1). Among the 6 schemes with at least 95% coverage, skill still spreads from 0.179 to 0.423, so coverage does not explain everything. Within that group, OpenAlex Topics scores best. It was built from research literature, the same kind of text as the corpus [22]. EuroVoc, a policy and legal thesaurus, scores lowest. That fits the expectation that a vocabulary matched to the corpus domain organizes it better. But Table 6 shows the classifier produces many low-confidence placements: in 9 of the 13 corpus runs, over 90% of placed items are flagged. The classifier's reach is itself part of the score (Section 5.3).

The holdout results are near zero for every scheme except `lcc-full` and `ilc2` (Table 5). On titles of 3.5 words, the lexical classifier finds a candidate for only 4.4% (BK) to 56.6% (complete LCC) of the books.

Table 4. Placed by shelfmark (`profile:lexical@2`) on the arXiv corpus.

scheme	cov.	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log ($a = 1$)
openalex-topics	97.3%	0.641	0.359	0.423	77.6	4.45	5	554	0.892	26.9
unesco-thesaurus	97.3%	0.721	0.279	0.329	146.7	4.90	9	535	0.852	26.7
ipc	98.0%	0.769	0.231	0.272	317.8	6.31	10	794	0.838	27.3
ilc2	100.0%	0.819	0.181	0.214	295.1	5.09	13	417	0.824	23.4
lcc-full	100.0%	0.823	0.177	0.209	657.2	6.06	11	121	0.803	22.9
acm-ccs	74.5%	0.831	0.169	0.200	84.9	3.61	7	877	0.861	23.7
fdc	77.0%	0.834	0.166	0.196	54.5	3.21	7	1103	0.836	25.6
eurovoc	97.4%	0.848	0.152	0.179	269.3	4.98	8	350	0.821	24.8
lcc	87.7%	0.849	0.151	0.178	129.0	4.08	10	661	0.837	25.4
arxiv	42.6%	0.891	0.109	0.128	14.5	2.21	4	1665	0.856	20.3
bk	48.1%	0.902	0.098	0.116	75.0	3.15	8	788	0.841	20.9
jel	47.9%	0.903	0.097	0.115	28.9	2.27	4	972	0.806	19.4
msc	39.2%	0.950	0.050	0.059	114.4	2.00	4	380	0.800	17.8

Table 5. Placed by shelfmark on the book holdout (general schemes only).

scheme	cov.	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log ($a = 1$)
lcc-full	56.6%	0.925	0.075	0.107	340.8	4.14	11	13	0.492	14.0
ilc2	45.6%	0.954	0.046	0.065	188.1	2.94	11	25	0.527	12.5
unesco-thesaurus	15.8%	0.989	0.011	0.016	93.8	1.65	7	8	0.439	10.8
lcc	8.4%	0.996	0.004	0.005	63.0	1.29	8	6	0.329	10.2
fdc	5.2%	0.998	0.002	0.003	31.1	1.14	7	4	0.314	10.1
bk	4.4%	0.999	0.001	0.002	30.8	1.23	7	7	0.439	10.1

Table 6. Placement statistics for the `lexical@2` runs. "Flagged" placements are below the 0.8 confidence target but kept. "Unplaced" counts fallbacks. "Top matched term" is the vocabulary term that sent items to the top class.

scheme	set	placed	flagged (<0.8)	unplaced	top class	top share	top matched term
acm-ccs	corpus	7,445	6,854	2,555	10002944.10011123.10011674	11.8%	performance
arxiv	corpus	4,260	3,325	5,740	cs.PF	39.1%	performance
bk	corpus	4,806	3,980	5,194	01.00	16.4%	general
eurovoc	corpus	9,737	9,582	263	C2228	3.6%	paper
fdc	corpus	7,696	7,023	2,304	676	14.3%	paper
ilc2	corpus	9,999	9,999	1	edqam	4.2%	s
ipc	corpus	9,799	9,725	201	G09B23/00	8.1%	models
jel	corpus	4,786	3,854	5,214	M53	20.3%	training
lcc	corpus	8,765	8,320	1,235	QA299.6-433	7.5%	analysis
lcc-full	corpus	9,999	9,999	1	QC243.3.E5	1.2%	energy
msc	corpus	3,916	2,999	6,084	51-XX	9.7%	geometry
openalex-topics	corpus	9,732	9,616	268	F21	5.7%	energy
unesco-thesaurus	corpus	9,731	9,529	269	c155	5.5%	time
bk	holdout	22	22	478	71.32	31.8%	man
fdc	holdout	26	26	474	750	11.5%	art
ilc2	holdout	228	226	272	edqam	11.0%	s
lcc	holdout	42	42	458	QH501-531	7.1%	life
lcc-full	holdout	283	283	217	K23	2.1%	s
unesco-thesaurus	holdout	79	77	421	c155	5.1%	time

5.3 How large is the classifier effect?

Table 7 compares gold and classifier placements for the same scheme on the same items and graph.

Where gold coverage is high (arXiv, LCC), the classifier loses most of the skill. For arXiv, skill falls from 0.786 to 0.128. The classifier places only 42.6% of the papers and sends 39.1% of those to `cs.PF` (Performance) on the word "performance" (Table 6). For the LCC outline on books, skill falls from 0.419 to 0.005. Where gold coverage is tiny (MSC, JEL), the classifier scores *higher* than the gold labels, only because it places more items. Neither direction measures scheme quality. In this run, the classifier, not the scheme, accounts for most of the differences in classifier-placed skill. The ranking in Table 4 is therefore a ranking

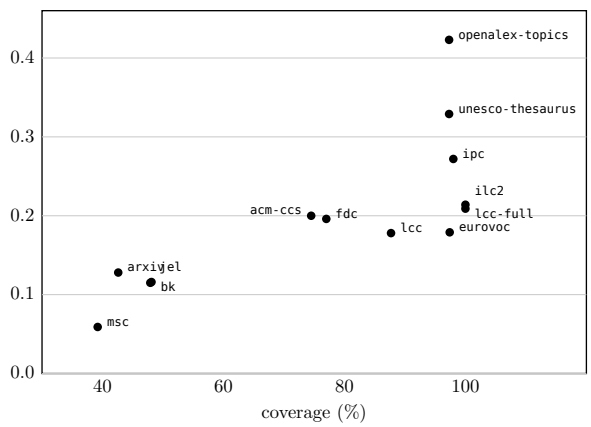


Figure 1. Coverage and skill for the 13 schemes placed by the lexical classifier on the arXiv corpus (data of Table 4).

Table 8. Depth and navigation cost under uniform use (items in the root menu included). a is the per-level overhead of Eq. 3.

scheme	labels	set	p50 depth	p95 depth	root fan-out	nav_log $a=0$	$a=1$	$a=2$	nav_scan
arxiv	gold	corpus	4	4	8	15.2	18.3	21.5	291
msc	gold	corpus	1	4	8738	13.9	15.2	16.5	4372
jel	gold	corpus	1	1	10000	13.3	14.3	15.3	5001
lcc	gold	holdout	4	6	53	14.8	18.6	22.4	69
lcc-full	gold	holdout	5	7	53	15.4	19.7	24.1	56
acm-ccs	placed	corpus	4	6	2568	20.5	23.7	27.0	1476
arxiv	placed	corpus	1	4	5748	18.5	20.3	22.2	3231
bk	placed	corpus	1	6	5199	18.3	20.9	23.5	2736
eurovoc	placed	corpus	5	6	284	20.2	24.8	29.4	269
fdc	placed	corpus	4	5	2314	22.4	25.6	28.7	1413
ilc2	placed	corpus	5	7	68	18.9	23.4	27.9	136
ipc	placed	corpus	6	9	209	22.3	27.3	32.3	322
jel	placed	corpus	1	4	5231	17.4	19.4	21.4	2876
lcc	placed	corpus	4	6	1255	21.7	25.4	29.0	825
lcc-full	placed	corpus	6	8	22	17.5	22.9	28.2	52
msc	placed	corpus	1	4	6141	16.0	17.8	19.7	3128
openalex-topics	placed	corpus	5	5	272	22.4	26.9	31.3	311
unesco-thesaurus	placed	corpus	5	7	276	22.0	26.7	31.4	305
bk	placed	holdout	1	1	481	9.1	10.1	11.2	242
fdc	placed	holdout	1	2	482	9.0	10.1	11.2	243
ilc2	placed	holdout	1	7	300	10.3	12.5	14.6	155
lcc	placed	holdout	1	4	470	9.1	10.2	11.3	237
lcc-full	placed	holdout	5	9	236	11.3	14.0	16.7	125
unesco-thesaurus	placed	holdout	1	6	427	9.4	10.8	12.2	216

Table 7. Gold against classifier placement on identical items and graph.

scheme	set	gold cov.	placed cov.	gold skill	placed skill	placed − gold
arxiv	corpus	100.0%	42.6%	0.786	0.128	−0.658
msc	corpus	13.2%	39.2%	0.018	0.059	+0.041
jel	corpus	0.02%	47.9%	0.000	0.115	+0.115
lcc	holdout	92.2%	8.4%	0.419	0.005	−0.413
lcc-full	holdout	92.2%	56.6%	0.421	0.107	−0.314

of *scheme as reached by this classifier*. We do not present it as a ranking of schemes. Section 8 controls coverage and the classifier tier and finds the same.

5.4 Navigation

The navigation models answer a different question from cost: how much reading it takes to reach an item, not whether related items are near each other. With unplaced items in the root menu, the two models diverge. $\mathrm{nav}_{\log}$ *drops* as coverage falls, because one large root menu costs only $\log_2 b$. $\mathrm{nav}_{\mathrm{scan}}$ rises steeply, because the root must be scanned linearly. Among high-coverage runs, the complete LCC schedules have the lowest scan cost ($\mathrm{nav}_{\mathrm{scan}}$ 52 on the corpus) and the most effective classes. This is the deep, narrow-shelf trade-off: more levels to descend, but short shelves at the bottom. The models' parameters are assumptions, so Table 8 shows the trade-off and does not settle it. The menu studies cited in Section 1.1 disagree on the best depth and breadth [45], [47], [48], so they do not fix the parameters either.

6 A METHODS NOTE: THE STOPWORD DEFECT

The first placed run (`profile:lexical@1`) showed single-class captures. The classifier's vocabulary tier matched complete class labels and terms, and some of them consist only of English function words. ILC2 has a class labelled "to", IPC a C01G catchword "TO" and the complete LCC a class `PL2813` labelled "To". Any abstract containing "to" matched them. The fix (classifier `vocab-2`, `profile:lexical@2`) drops normalized labels and terms whose tokens are all stopwords. Labels with at least one content word still match in full. We then reran only the placed part, on the same graphs, scheme files (byte-identical, checked by SHA-256) and reference costs. The gold results do not depend on the classifier and were not rerun on this account. Every placed number in this paper comes from the rerun.

The fix removed the captures (Table 9). ILC2's largest class fell from 73.7% to 4.2% of placed papers, IPC's from 60.1% to 8.1%, complete LCC's from 22.1% to 1.2% and the UNESCO Thesaurus's from 17.8% to 5.5%. Skill moved much less. On the corpus, 6 of 13 schemes changed skill by at least 0.0005, and Kendall's τ_b between the skill values before and after is 0.95. Complete LCC *lost* skill (0.236 → 0.209) even though its placements became plausible. A plausible explanation, not tested here, is that the wrong placements had piled most papers that share "to" into a single class. Within such a pile, the edges the graph keeps still connect similar papers, and the pile is far smaller than the whole corpus, so its edges cost less than they would under the root. A metric can reward a degenerate placement more than a spread of honest but noisy placements. This is a known property of a cost that charges same-class pairs the class size: one large class still beats the root. It is also why we always report largest-class share and placement statistics next to skill.

7 VALIDITY OF THE COST

The cost scores agreement with a similarity graph. Whether it tracks what a librarian would call a good decision is a separate question, and we test it rather than assume it.

Table 9. Effect of the stopword fix (before: `lexical@1`; after: `lexical@2`). "Top class" gives the largest single-class share of placed items and its code.

scheme	set	top class before	top class after	coverage	cost_norm	skill	Δ skill
acm-ccs	corpus	11.8% 10002944.10011123.10011674	11.8% 10002944.10011123.10011674	74.5% → 74.5%	0.831 → 0.831	0.200 → 0.200	0.000
arxiv	corpus	39.1% cs.PF	39.1% cs.PF	42.6% → 42.6%	0.891 → 0.891	0.128 → 0.128	0.000
bk	corpus	16.4% 01.00	16.4% 01.00	48.1% → 48.1%	0.902 → 0.902	0.116 → 0.116	0.000
eurovoc	corpus	3.6% C2228	3.6% C2228	97.4% → 97.4%	0.848 → 0.848	0.179 → 0.179	+0.001
fdc	corpus	14.3% 676	14.3% 676	77.0% → 77.0%	0.834 → 0.834	0.196 → 0.196	0.000
ilc2	corpus	73.7% agy993	4.2% edqam	100.0% → 100.0%	0.825 → 0.819	0.206 → 0.214	+0.007
ipc	corpus	60.1% C01G	8.1% G09B23/00	99.4% → 98.0%	0.781 → 0.769	0.259 → 0.272	+0.014
jel	corpus	18.7% M53	20.3% M53	51.7% → 47.9%	0.902 → 0.903	0.116 → 0.115	−0.001
lcc	corpus	7.5% QA299.6-433	7.5% QA299.6-433	88.4% → 87.7%	0.848 → 0.849	0.179 → 0.178	0.000
lcc-full	corpus	22.1% PL2813	1.2% QC243.3.E5	100.0% → 100.0%	0.800 → 0.823	0.236 → 0.209	−0.027
msc	corpus	9.7% 51-XX	9.7% 51-XX	39.2% → 39.2%	0.950 → 0.950	0.059 → 0.059	0.000
openalex-topics	corpus	5.7% F21	5.7% F21	97.4% → 97.3%	0.641 → 0.641	0.423 → 0.423	0.000
unesco-thesaurus	corpus	17.8% c493	5.5% c155	98.9% → 97.3%	0.732 → 0.721	0.316 → 0.329	+0.013
bk	holdout	31.8% 71.32	31.8% 71.32	4.4% → 4.4%	0.999 → 0.999	0.002 → 0.002	0.000
fdc	holdout	11.5% 750	11.5% 750	5.2% → 5.2%	0.998 → 0.998	0.003 → 0.003	0.000
ilc2	holdout	13.7% agqs99	11.0% edqam	56.8% → 45.6%	0.934 → 0.954	0.095 → 0.065	−0.029
lcc	holdout	7.1% QH501-531	7.1% QH501-531	8.4% → 8.4%	0.996 → 0.996	0.005 → 0.005	0.000
lcc-full	holdout	3.9% BQ940	2.1% K23	61.0% → 56.6%	0.917 → 0.925	0.119 → 0.107	−0.011
unesco-thesaurus	holdout	5.0% c155	5.1% c155	16.0% → 15.8%	0.989 → 0.989	0.016 → 0.016	0.000

Table 11. Judge agreement before and after adding the false-join term, per run.

run	⋉, C [95% CI]	⋉, C + FJs [95% CI]	change in ⋉
arxiv-gold	0.24 [0.05, 0.43]	0.11 [0.05, 0.18]	-0.13 [-0.28, +0.04]
msc-gold	0.20 [0.01, 0.38]	0.33 [0.21, 0.46]	+0.13 [-0.07, +0.33]
fdc-placed	0.29 [0.21, 0.38]	0.63 [0.48, 0.78]	+0.34 [+0.19, +0.47]
ilc2-placed	0.25 [0.18, 0.33]	0.66 [0.50, 0.79]	+0.40 [+0.28, +0.50]
lcc-full-placed	0.25 [0.19, 0.32]	0.68 [0.54, 0.81]	+0.43 [+0.33, +0.52]
openalex-topics-placed	0.26 [0.16, 0.35]	0.66 [0.52, 0.79]	+0.40 [+0.28, +0.50]
unesco-thesaurus-placed	0.24 [0.17, 0.32]	0.54 [0.38, 0.69]	+0.30 [+0.16, +0.43]
learned-placed	0.33 [0.16, 0.50]	0.30 [0.18, 0.43]	-0.03 [-0.20, +0.14]

7.1 A blind judge, and a false-join term

We drew 953 pairs from eight runs in three strata: similar pairs the scheme splits at depth one or less (A), dissimilar pairs it joins deep (B), and random placed pairs (C). One LLM judge labelled each pair blind under a rubric fixed before judging (`docs/validity`). The metric flags a pair when its cost contribution $w\ |\text{leaves(lca)}|\ /n$ exceeds the run's mean contribution per graph edge. We report Cohen's κ [83] and the AUC of the raw contribution, with percentile intervals that resample pairs within strata [84]. Over all pairs, the flag agrees with the judge at $\kappa = 0.27$ [0.22, 0.32], with agreement 0.63 and AUC 0.65. That is weak agreement. The judge is stable against itself ($\kappa = 0.92$ on a blind second pass of 238 pairs).

The strata show where the gap is. Among similar pairs the scheme splits (A), the judge calls 0.70 incorrect and the metric flags 0.84. Among dissimilar pairs the scheme joins deep (B), the judge also calls 0.70 incorrect and the metric flags 0.00: those pairs have zero weight, so the cost is blind to them by construction. Shares describe a stratified sample, one third per stratum, not the population of pairs. Over the eight runs, the Spearman correlation of a run's `cost_norm` with the judge's error rate on random pairs is 0.23, and over all strata 0.42 with eight points this is descriptive only.

The correction adds a false-join term. For a seeded sample of non-edge pairs whose cosine is below τ, weighted by dissimilarity $d = \tau - \cos$, the specificity form is

$$\text{FJs} = \frac{\sum d \ln(n/N(\text{lca}))}{\ln n \sum d}, \tag{5}$$

so that deep joins of dissimilar items cost more than shallow ones. The corrected objective is $J' = C + \beta\ \text{FJs} + \lambda N$ with $\beta = 1$, $\lambda = 0.002$ and N the $\text{nav}_{\log}$ cost in bits. An outside-mass form is also computed. Adding the term raises agreement to $\kappa = 0.48$ [0.43, 0.53], a paired change of +0.21 [+0.15, +0.27], and the AUC from 0.65 to 0.78 (Table 10). The term alone does not do it ($\kappa = 0.21$). Weights from 0.25 to 8 give κ between 0.45 and 0.49, so β is not critical in that range. Every placed run except the learned scheme gains, with intervals that exclude zero; arXiv's own taxonomy and the learned scheme do not, because the judge rarely finds a false join in them (Table 11).

7.2 Checks that do not use the judge

Two signals are independent of the LLM judge: a shared Library of Congress subject heading between books, and

Table 10. Agreement of the metric's flag with the judge, pooled over all pairs, for Dasgupta cost and the false-join variants (same pairs and labels).

variant	flagged	⋉ [95% CI]	AUC [95% CI]
Dasgupta cost C	0.28	0.27 [0.22, 0.32]	0.65 [0.62, 0.67]
FJs alone	0.34	0.21 [0.15, 0.26]	0.61 [0.59, 0.64]
C + 0.5 FJs	0.60	0.49 [0.44, 0.55]	0.80 [0.77, 0.83]
C + FJs	0.62	0.48 [0.43, 0.53]	0.78 [0.75, 0.81]
C + 4 FJs	0.65	0.45 [0.40, 0.51]	0.76 [0.73, 0.79]
C + FJ (outside mass)	0.61	0.47 [0.41, 0.52]	0.78 [0.75, 0.81]

Table 12. Agreement of the cost and the corrected cost with non-LLM signals. Separation: difference in AUC between the FDC and LCC backbones, with a 95% interval from paired draws.

external signal	$\varkappa$, C	$\varkappa$, C + FJs	scheme-level ϱ, C	scheme-level ϱ, C + FJs	FDC-backbone minus LCC-backbone separation [95% CI]
shared LCSH heading (books)	0.24	0.21	0.88	0.89	ddc-section - lcc-gold: -0.002 [-0.011, +0.007]
bibliographic coupling (papers)	0.13	0.06	0.90	0.90	fdc-arxiv-nested - lcc-arxiv-nested: +0.004 [-0.001, +0.010]
co-citation (papers)	0.23	0.19	0.87	0.87	fdc-arxiv-nested - lcc-arxiv-nested: +0.001 [-0.026, +0.025]

Table 13. Lexical (`profile:lexical@2`) and shortlist (`profile:shortlist@2`) placement on the arXiv corpus.

scheme	coverage lexical	coverage shortlist	skill lexical	skill shortlist	change in skill
acm-ccs	0.745	1.000	0.200	0.278	+0.078
arxiv	0.426	1.000	0.128	0.502	+0.373
bk	0.481	1.000	0.116	0.360	+0.244
eurovoc	0.974	1.000	0.179	0.175	-0.004
fdc	0.770	1.000	0.196	0.258	+0.062
ilc2	1.000	1.000	0.214	0.244	+0.031
ipc	0.980	1.000	0.272	0.296	+0.023
jel	0.479	1.000	0.115	0.303	+0.189
lcc	0.876	1.000	0.178	0.212	+0.033
lcc-full	1.000	1.000	0.209	0.217	+0.008
msc	0.392	1.000	0.059	0.230	+0.171
openalex-topics	0.973	1.000	0.423	0.465	+0.042
unesco-thesaurus	0.973	1.000	0.329	0.338	+0.009

bibliographic coupling or co-citation between arXiv papers from OpenAlex reference lists. Table 12 gives the pair-level agreement (κ) of each flag with the signal, the scheme-level rank agreement (ρ) and the separation each signal finds between an FDC backbone and an LCC backbone. The false-join term does *not* raise pair-level agreement with any of the three (coupling 0.13 to 0.06, co-citation 0.23 to 0.19, subject headings 0.24 to 0.21), and leaves the scheme-level rank agreement where it was. The corrected metric is better supported than the original by one annotator model, and not by independent signals. These signals have their own limits: most 2025 to 2026 preprints in the corpus have no OpenAlex reference list, and LC assigns subject headings and call numbers in one record, so that signal is not independent of LCC.

The judge's together-or-apart call agrees with whether the pair is an edge of the graph at $\kappa = 0.62$, and disagrees with whether the scheme put the pair together (κ = -0.31). On its sample the judge finds a false-join share of 0.24 and a false-split share of 0.27 of all sampled pairs, shares that depend on how the strata were drawn.

8 COVERAGE, CLASSIFIER TIER AND RANK STABILITY

8.1 Coverage control

A gold run's unlabelled items hang under the root and are charged the maximum, so coverage enters the cost directly. The covered-subset analysis rescored the placed runs on each run's own covered items. Coverage explains 59% of the rank variance of Dasgupta cost over the 14 placed runs on the papers (ρ = -0.77), and the ranking by cost on all items and on each run's own covered items barely agrees (Kendall $\tau = 0.10$). Low-coverage schemes move from the bottom to the middle once their unplaced papers are dropped. The sparse graph is not the cause: on random subsets of 1,000 to 5,000 papers, the kNN-sparse cost and a dense cost rank the placed runs with Spearman $\rho \geq 0.92$.

8.2 A second classifier tier

The shortlist profile adds a BGE-small embedding shortlist after the lexical tiers and files every item, so coverage is 1.000 for all 13 schemes. The rerank profile is unavailable in this build and reproduces the shortlist placements. Table 13 compares the tiers on the same graph and schemes. Skill rises for 12 of 13 schemes, by much more where the lexical tier placed few papers. arXiv's own taxonomy moves from 0.128 to 0.502 and now leads, with `openalex-topics` second (0.465), and the Spearman correlation between the two tiers' skill values is 0.14. The order of schemes under a classifier is a property of the classifier–scheme pair. The run's directory is marked pending in the results index, so the numbers are read from its committed JSON and should be treated as preliminary. The tier that places by embedding also uses a different model from the graph's.

8.3 Rank stability on the papers

The stability sweep rescored the 13 schemes under 54 controlled settings: three embedding models, k of 10, 20 and 50, three graph thresholds, and the lexical and shortlist tiers. A ranking claim is called robust when the minimum Spearman ρ over every pair of settings is at least 0.8. Table 14 gives the minimum and median ρ between settings that use the same tier and between settings that do not.

Within a tier the rankings are stable: cost, revenue and skill have a minimum ρ of 0.90 over the embedding, k and threshold sweep. Across tiers they are not: the minimum is 0.04 and the median 0.15. The leader under cost is `openalex-topics` under the lexical tier and `arxiv` under the shortlist tier. Objectives also disagree. FDC ranks above LCC by cost, revenue, skill and locality in 0 settings. By triplet

Table 14. Rank agreement of the 13 schemes between controlled settings, by objective, and the number of settings in which FDC ranks above LCC.

objective	min ρ, same tier	median ρ, same tier	min ρ, across tiers	median ρ, across tiers	FDC above LCC (settings)
Dasgupta cost	0.896	0.978	0.038	0.154	0 of 54
Moseley-Wang revenue	0.896	0.978	0.038	0.154	0 of 54
skill	0.896	0.978	0.038	0.154	0 of 54
triplet accuracy	0.808	0.984	-0.280	-0.104	27 of 54
shelf-order locality	0.780	0.962	0.549	0.736	0 of 54

Table 15. Books, MiniLM, $k = 20$, published graph. C is `cost_norm`; FJs enters at $\beta = 1$; J' adds $0.002N$.

run	structure	coverage	C	C + FJs	nav bits	J'	skill
`ddc-gold`	Dewey, librarian number, full depth	1.000	0.493	0.520	19.3	0.558	0.658
`ddc-section`	Dewey, cut to section (3 digits)	1.000	0.494	0.521	17.1	0.555	0.657
`fdc-d3`	FDC from Dewey, depth 3	1.000	0.496	0.523	17.7	0.559	0.654
`fdc-d2`	FDC from Dewey, depth 2	1.000	0.502	0.528	16.1	0.561	0.647
`lcc-gold`	LCC, librarian call number (outline)	1.000	0.523	0.531	16.2	0.563	0.619
`lcc-full-gold`	LCC, librarian call number (complete schedules)	1.000	0.522	0.530	16.8	0.563	0.620
`learned-own`	learned scheme, own insertion	1.000	0.339	0.391	29.7	0.450	0.858
`fdc-placed`	FDC, lexical classifier	0.839	0.654	0.680	23.4	0.727	0.450
`lcc-lcsh-placed`	LCC outline with LCSH vocabulary, lexical classifier	0.985	0.674	0.689	18.6	0.726	0.423
`lcc-placed`	LCC outline, lexical classifier	0.915	0.733	0.750	20.6	0.791	0.346
`unesco-thesaurus-placed`	UNESCO Thesaurus, lexical classifier	0.925	0.629	0.656	22.2	0.700	0.482

Table 16. Pair claims across 27 settings. "Claimed" is the sign the proposal reported; "sign matches" counts settings with that sign; "significant" counts settings whose 95% interval excludes zero, for and against the claim. A tie claim fails when any interval excludes zero.

comparison (a vs b)	objective	claimed	sign matches	significant for / against	range of a - b	verdict
Dewey sections vs LCC, librarian numbers	C	a < b	13 of 27	10 / 9	[-0.0753, +0.0750]	not stable
Dewey sections vs LCC, librarian numbers	J'	tie	8 of 27	- / 19	[-0.0564, +0.0948]	not stable
Dewey sections vs FDC depth 3	C	a < b	27 of 27	27 / 0	[-0.0044, -0.0010]	stable
FDC depth 3 vs depth 2	J'	a < b	27 of 27	24 / 0	[-0.0085, -0.0001]	stable
FDC depth 3 vs full FDC number	J'	a < b	15 of 27	9 / 1	[-0.0005, +0.0004]	not stable
Full Dewey number vs section	J'	a > b	27 of 27	27 / 0	[+0.0013, +0.0036]	stable
Learned scheme vs Dewey sections	J'	a < b	19 of 27	13 / 6	[-0.1520, +0.0917]	not stable
Placed: FDC vs LCC outline	J'	a < b	27 of 27	24 / 0	[-0.0897, -0.0237]	stable
Placed: FDC vs LCC outline + LCSH	J'	a < b	9 of 27	5 / 8	[-0.0434, +0.0729]	not stable

accuracy it ranks above LCC in 27 of 54, and that ranking can reverse between settings (minimum ρ -0.28). A claim that one scheme beats another on these papers therefore depends on the classifier and on the objective. The graph settings matter much less.

9 BOOKS WITH DEWEY NUMBERS: DEWEY AGAINST LCC

A third file holds 3,000 Open Library editions that each carry an LC call number, a Dewey number and subjects (320 also have a description). The 21 LCC main classes have equal quotas, and no edition shares a work with the 500-book holdout. The graph text is the description followed by up to 30 subject headings, so it is mostly Library of Congress subject headings, and $\tau = 0.157$. Dewey numbers are used as notation only: the tree is the digit prefixes, and no Dewey caption, schedule or table is read (Section 12). Every item has both a Dewey number and an LC call number from librarians, so the Dewey and LCC runs are gold on the same items and coverage is 1.000 for both. Table 15 gives the runs on the MiniLM $k = 20$ graph. Table 16 gives the comparisons that a book-structure proposal drew from them, rechecked in each of 27 settings (three embedding models, three values of k and three graph thresholds, with 1,000 paired item-bootstrap draws each).

Dewey sections beat LCC on cost on the published graph for all three embedding models (-0.0292 [-0.0438, -0.0141] on MiniLM, -0.0411 on bge-small, -0.0239 on nomic, at $k = 20$). Across the 27 settings the sign matches in 13, with 10 significant for and 9 significant against, and the difference ranges from -0.0753 to +0.0750. Under the corrected objective the claim becomes a tie, which holds in 8 settings with 19 significant departures. The worst-case regret that earlier favoured Dewey sections (0.060 against 0.067 for LCC over the nine published graphs) reverses over all 27 settings (0.116 against 0.067). The non-LLM signals do not separate them either (Section 7.2): shared subject headings -0.002 [-0.011, +0.007], and for papers shelved under an FDC or an LCC backbone, coupling +0.004 [-0.001, +0.010] and co-citation +0.001 [-0.026, +0.025]. We read this as a tie that depends on the setting. The sample is stratified equally over LCC's main classes, which favours LCC's false-join term, and its text is mostly LC subject headings.

Smaller comparisons are stable in every setting. A three-digit Dewey section beats FDC depth 3 by less than 0.0044 of cost, a caption-versus-number difference. Three levels beat two. The full Dewey number gains cost and loses on J'. Through the same classifier, FDC beats the LCC outline without subject-heading vocabulary in all 27 settings. It does not beat the bundled LCC outline with LCSH vocabulary, which places 0.99 of the books against

Table 18. The fitted scheme against published schemes on the same graphs. "own, in-sample" rows are the fitted assignment of the corpus it was fitted to. "own insertion" rows file works by nearest leaf centroid in the embedding space of the graph. Rows through `lexical@2` use the classifier of Section 5.2. Lower is better for cost and J.

evaluation set	scheme	placements	coverage	cost_norm	skill	J	nav_log ($a = 1$)
corpus, n = 10,000	learned	own, in-sample (fitted on this corpus)	1.000	0.172	0.976	0.264	18.5
corpus, n = 10,000	arxiv	gold (source labels)	1.000	0.333	0.786	0.425	18.3
corpus, n = 10,000	learned	shelfmark lexical@2	0.999	0.379	0.732	0.535	31.1
corpus, n = 10,000	openalex-topics	shelfmark lexical@2	0.973	0.641	0.423	0.776	26.9
corpus, n = 10,000	lcc-full	shelfmark lexical@2	1.000	0.823	0.209	0.937	22.9
held-out 20%, n = 2,000	learned (fit on the other 80%)	own insertion, held out	1.000	0.172	0.922	0.253	16.1
held-out 20%, n = 2,000	learned (full fit)	own, in-sample	1.000	0.179	0.914	0.260	16.2
held-out 20%, n = 2,000	arxiv	gold (source labels)	1.000	0.342	0.733	0.422	15.9
held-out 20%, n = 2,000	openalex-topics	shelfmark lexical@2	0.975	0.652	0.388	0.758	21.2
book holdout, n = 500	learned	own insertion (embedding nearest centroid)	1.000	0.443	0.797	0.542	19.7
book holdout, n = 500	lcc	gold (source labels)	0.922	0.708	0.419	0.801	18.6
book holdout, n = 500	lcc-full	gold (source labels)	0.922	0.706	0.421	0.804	19.7
book holdout, n = 500	lcc-full	shelfmark lexical@2	0.566	0.925	0.107	0.995	14.0
book holdout, n = 500	learned	shelfmark lexical@2	0.086	0.992	0.012	1.044	10.6

0.84 for FDC: that comparison turns on vocabulary and coverage, not on the scheme. On the 500-book title-only holdout no placed run reaches the coverage floor of 0.80, and the best reaches 0.57.

The default scheme. The tool needs one default scheme for a general library of mostly books, filed by its classifier alone. Its decision record states a rule: among placed runs whose coverage is at least a floor, take the highest skill; every run whose paired difference to it has a 95% interval covering zero is tied; the tied run with the lowest navigation cost wins. The floor is 0.90, so that at most one book in ten waits for a person. Table 17 gives the winners in the 27 settings. At the 0.90 floor the LCC outline with LCSH headings as search vocabulary (`lcc-lcsh`) wins all 27 of 27 settings among the open schemes the tool may bundle (FDC, the LCC outline with and without LCSH vocabulary, LCC full, BK and the fitted scheme), and it is the only winner at every floor from 0.85 up. Over all placed schemes it wins 15 and the UNESCO Thesaurus, which may not be bundled, 12. Part of the win is vocabulary: LCSH lifts the outline's coverage from 0.915 to 0.985, while FDC places 0.839 with its captions alone. The bundled default of the tool is this scheme. The decision record also documents a catalogue-first alternative, FDC cut to Dewey sections, which rests on the gold-number tie above and on openness and notation length rather than on a metric result.

Table 17. Winner of the default-scheme rule on the books: settings won out of 27, by coverage floor, over all placed schemes and over the open schemes the tool may bundle.

coverage floor	winner, all placed schemes (settings)	winner, bundleable open schemes (settings)
0.70	lcc-lcsh-placed 15, unesco-thesaurus-placed 12	lcc-lcsh-placed 18, fdc-placed 9
0.80	lcc-lcsh-placed 15, unesco-thesaurus-placed 12	lcc-lcsh-placed 18, fdc-placed 9
0.85	lcc-lcsh-placed 15, unesco-thesaurus-placed 12	lcc-lcsh-placed 27
0.90	lcc-lcsh-placed 15, unesco-thesaurus-placed 12	lcc-lcsh-placed 27
0.95	lcc-lcsh-placed 27	lcc-lcsh-placed 27

10 A FITTED ORDER-PRESERVING SCHEME

The benchmark makes a converse question natural: if a scheme can be scored, can one be *fitted* to the score, under constraints a library needs? We report a first, short experiment; the full description is in a companion draft.

Objective. On the graph of Section 3.1, the fitted scheme minimizes

$$J(T) = \text{cost}_{\text{norm}}(T) + \lambda N(T), \quad \lambda = 0.005, \qquad (6)$$

where $\text{cost}_{\text{norm}}$ is the harness's cost with works under their class and N is $\text{nav}_{\log}$ with $a = 1$ (Eq. 3; a leaf class's fan-out is its number of works). The constraints are 2 to 12 child classes per class, at most 4 class levels, leaf classes of at least 5 works, and every class a contiguous run of one fixed shelf order, the alphabetic-tree constraint of optimal order-preserving codes [42], [85]. New works are filed by splitting leaf classes only, so that every existing class keeps its code, label and parent. The fit starts from the average-linkage tree, collapses it to a bounded-shape class tree by an exact dynamic program, and applies order-preserving local search; it takes about 2.2 s on the corpus. The result has 110 classes (2, 5, 16, 87 by level). Their labels were written by hand from sheets of class terms and central titles, by the AI coding assistant that built the prototype and not by a cataloguer.

Results. In sample, the fitted scheme reaches cost 0.1722 (average linkage 0.1518, arXiv's own categories 0.333), skill 0.9760 and J 0.2644 (Table 18). Fitted on a random 80% and filed on the other 20% by insertion, it scores 0.1722 on the test works, against 0.1793 for the full fit that saw them. Reached through the lexical classifier, it scores 0.3794: a lower cost than every published scheme's placed row, and a higher cost than arXiv's gold labels. Inserting the 500 books as new works produced 16 classes and changed no existing identifier on the holdout graph the inserted scheme scores 0.4432 (skill 0.7971), against 0.708 for LCC call numbers. Through the classifier it places only 0.086 of the books and scores 0.9915.

Caveats. These numbers flatter the method, and we do not read them as a win over any published scheme.

Table 19. The fitted scheme against the best other row in each of 9 graph conditions. J is the fitted objective.

comparison	learned first by skill	skill margin CI > 0	J margin CI > 0	tightest condition: skill margin [95% CI]
held out 20% (MiniLM fit, inserted)	9 of 9	9 of 9	9 of 9	bge-small-k10: 0.155 [0.122, 0.189]
in sample (full MiniLM fit)	9 of 9	9 of 9	9 of 9	bge-small-k50: 0.160 [0.151, 0.169]
500 holdout books, inserted	9 of 9	9 of 9	9 of 9	nomic-k10: 0.280 [0.214, 0.340]
placed by lexical@2	9 of 9	9 of 9	9 of 9	nomic-k10: 0.290 [0.277, 0.302]

Table 20. Fitted scheme minus the best non-learned run under J' and under $C + \mathrm{FJ}$, both at $\beta = 1$, with paired 95% intervals over 1,000 item draws. Negative means the fitted scheme is ahead.

universe	model	k	best other by J'	J' learned - best [95% CI]	best other by C + FJ	C + FJ learned - best [95% CI]
3,000 books	minilm	10	lcc-gold	-0.016 [-0.043, +0.007]	lcc-gold	+0.140 [+0.111, +0.167]
3,000 books	minilm	20	ddc-section	-0.105 [-0.129, -0.083]	lcc-full-gold	+0.045 [+0.020, +0.068]
3,000 books	minilm	50	ddc-d2	-0.152 [-0.171, -0.134]	lcc-gold	-0.037 [-0.059, -0.016]
3,000 books	bge-small	10	lcc-gold	+0.018 [-0.008, +0.040]	lcc-gold	+0.183 [+0.155, +0.209]
3,000 books	bge-small	20	ddc-section	-0.052 [-0.075, -0.032]	lcc-full-gold	+0.092 [+0.068, +0.115]
3,000 books	bge-small	50	ddc-d2	-0.097 [-0.116, -0.079]	ddc-gold	+0.015 [-0.007, +0.038]
3,000 books	nomic	10	lcc-gold	+0.036 [+0.010, +0.060]	lcc-gold	+0.195 [+0.168, +0.223]
3,000 books	nomic	20	ddc-section	-0.058 [-0.082, -0.036]	lcc-full-gold	+0.098 [+0.072, +0.123]
3,000 books	nomic	50	ddc-section	-0.108 [-0.129, -0.090]	lcc-gold	+0.010 [-0.012, +0.033]
10,000 papers (in sample)	minilm	20	fdc-arxiv-nested	-0.136 [-0.145, -0.128]	arxiv-gold	-0.041 [-0.051, -0.031]
10,000 papers (in sample)	bge-small	20	fdc-arxiv-nested	-0.115 [-0.122, -0.106]	arxiv-gold	-0.046 [-0.056, -0.036]
10,000 papers (in sample)	nomic	20	fdc-arxiv-nested	-0.116 [-0.124, -0.108]	arxiv-gold	-0.036 [-0.047, -0.027]
500 holdout books (title only)	minilm	20	lcc-pipeline	-0.264 [-0.309, -0.221]	lcc-full-gold	-0.249 [-0.306, -0.192]
500 holdout books (title only)	bge-small	20	fdc-pipeline	-0.181 [-0.226, -0.137]	fdc-work	-0.110 [-0.168, -0.053]
500 holdout books (title only)	nomic	20	fdc-work	-0.184 [-0.235, -0.126]	fdc-work	-0.134 [-0.198, -0.066]

- *In sample.* The corpus rows, and the classifier rows of the fitted scheme, use a tree or a vocabulary fitted on those same works. The fitted scheme's vocabulary is the corpus's own most distinctive terms.
- *Embedding circularity.* Insertion files a new work at the nearest leaf centroid in the same all-MiniLM-L6-v2 space from which the graph is built, so a new work is placed by exactly the geometry it is later scored on. This holds for the held-out rows and, more strongly, for the book holdout, where graph and insertion see only title embeddings. The classifier rows are the comparison without it. Rescoring on graphs from two independent embedding models (below) removes the circularity of the scoring graph, not of filing.
- *One corpus.* The scheme has seen only arXiv papers. Its top level splits computing from mathematics and the physical sciences, and it has no class for the humanities until insertion creates a weak one. Labels of inserted classes come from frequent title words and are not usable as captions.
- *The objective.* Dasgupta cost favours a binary split near the top, so the fitted root has two children. Library schemes have wide top levels for reasons of browsing that J does not model.
- *Weights are modelling choices.* λ, a and the depth bound are not fitted. A sweep over depth bounds 3, 4, 6 and λ from 0.001 to 0.05 moves $\mathrm{cost}_{\mathrm{norm}}$ from 0.1577 to 0.2743 and the class count from 330 down to 15.

Home advantage. The fitted scheme's lead is largest on the graph it was fitted on. On the books of Section 9, its cost is lower than that of Dewey sections on every graph, but the margin shrinks off the fit graph (Table 21). Refits on the other two embedding models score best on their own graph (Table 22): on the held-out split, a fit scored on its own model's graph averages 0.926 skill against 0.904 elsewhere, and it is best on its own graph for 3 of 3 models. In the sample, the held-out and the classifier-placed rows read 0.976, 0.922 and 0.732 skill. The claim that the fitted scheme beats Dewey sections is not stable across the 27 book settings: the sign matches in 19 under J', with 6 significant reversals (Table 16). We use the fitted scheme as a reference for how low a tree can go on a graph, not as a candidate scheme.

Independent embeddings and false joins. Rescored on graphs from the two other embedding models at $k = 10$, 20 and 50, the fitted scheme still ranks first by skill in all 9 conditions, for the held-out split, the in-sample fit, the inserted book holdout and the classifier-placed run, with paired intervals of the margin above zero (Table 19). The tightest case is the held-out split on bge-small-k10, a margin of 0.155 [0.122, 0.189]. Under the corrected objective J', against the best non-learned run of each universe on the false-join rescoring's graphs (Table 20), it leads on the papers in 3 of 3 graphs and on the title-only holdout in 3 of 3. On the 3,000 books it leads in 6 of 9, ties in 2 and is behind in 1 under the outside-mass form it is behind in 6 of 9. Its edge survives a change of embedding. On books it does not survive the false-join correction.

Table 21. Fitted scheme against Dewey sections on books, by graph ($k = 20$). The scheme was fitted on MiniLM.

graph (books, k = 20)	C learned	C Dewey sections	C difference	J' difference
minilm (fit graph)	0.339	0.494	-0.155	-0.105
bge-small	0.376	0.483	-0.108	-0.052
nomic	0.361	0.473	-0.112	-0.058

Table 22. Held-out skill of each fit on each model's $k = 20$ papers graph.

learned fit	scored on minilm	scored on bge-small	scored on nomic
fitted on minilm	0.922	0.901	0.891
fitted on bge-small	0.913	0.938	0.909
fitted on nomic	0.899	0.912	0.918

11 DISCUSSION

What the measurement supports. The method gives one scale for schemes of different size, depth and domain. It costs little: the average-linkage reference took 58 s on the corpus graph and 0.1 s on the holdout graph, and no scoring run took more than 6.9 s. Its inputs are pinned by hash. On gold labels it separates a well-fitted scheme (arXiv on arXiv) from a general scheme on books (LCC on the holdout) and from schemes with too little gold to judge (MSC, JEL). Within one classifier, the order of schemes on the papers does not depend on the embedding model, k or graph threshold (Section 8.3). Several book-side comparisons are stable in every setting we tried: the cut depth of a Dewey-derived scheme, and the full number against its three-digit section (Section 9).

What it does not support. It does not support a ranking of universal schemes, for four reasons that each have a number behind them.

1. *Coverage.* Classifier placement is the only way to reach most schemes, and coverage explains 59% of the rank variance of cost over the placed runs (Section 8.1). Gold placement does not escape this: the MSC and JEL gold scores measure how few papers carry a code.
2. *The classifier.* The order of schemes changes when the classifier does: the Spearman correlation between the lexical and shortlist tiers' skill is 0.14, and the leader changes from openalex-topics to arxiv (Section 8.2, Section 8.3). The stopword defect shows the same risk from the other side: a classifier defect can change placements drastically while skill barely moves, or moves it the wrong way (Section 6).
3. *The objective.* FDC ranks above LCC under triplet accuracy in 27 of 54 settings and under cost in 0. Dewey sections against LCC on books is a setting-dependent tie: 13 of 27 settings match the claimed sign under cost, and 8 the tie under the corrected objective.
4. *The metric's validity.* The cost agrees with a blind judge at $\kappa = 0.27$. The false-join correction raises that to 0.48 against the judge and not against citation or subject-heading signals, and the judge is one model (Section 7).

A scheme's rank on this metric is therefore a statement about a similarity graph, a classifier and an objective. It is not a statement about the scheme's quality in use. The default scheme follows a stated rule rather than a ranking: the LCC outline with LCSH vocabulary wins it in 27 of 27 book settings among the open schemes, and the win is partly a vocabulary effect (Section 9). On librarian-assigned numbers the metrics do not separate Dewey from LCC.

Toward a fair comparison. The benchmark design lays out controls. A coverage control, a second classifier tier, an embedding, k and threshold sweep, item bootstrap intervals and a judge-based validity check have now been run. A permutation null that keeps each scheme's shape and class-size histogram, a label-intrinsic track that needs no works, a noise-injection curve that calibrates classifier error on LCC, and placement by a classifier other than shelfmark's (Annif or a prompted LLM) have not. The largest remaining gain would come from a second annotator model and a human annotator on the validity sample, and from a graph that is not built from embeddings (co-use or citations).

Interpretation of skill. Skill compares a scheme with a tree fitted to the same similarity graph. A scheme encodes many relations besides textual similarity: discipline, form, audience, place, period. A low skill can mean the scheme organizes by something the embedding does not see. Our score measures agreement with one notion of relatedness. It does not measure quality in every sense that matters to libraries.

On the fitted scheme. A scheme fitted to the benchmark's own objective scores well on that objective, as it must. The result is a floor for how low the cost can go on a given graph, not a ranking: its lead is partly home advantage and is not stable across settings (Section 10). Whether an open, learned, order-preserving classification can become a general one is open.

12 LIMITATIONS

- **Classifier bias.** Every non-gold placement comes from shelfmark's own classifier: the lexical profile for the main tables and a BGE-small shortlist profile for Section 8.2. The lexical profile matches class labels and attached vocabulary, so a scheme with rich vocabulary is reached better than one with bare labels, independent of its structure. Most placements are below the classifier's own confidence target (Table 6). Table 7 and Section 8.2 show the classifier effect is as large as the effects we want to measure. No classifier from outside this project (for example Annif or an LLM) was run for comparison. The semantic tier's directory is marked pending in the results index, so those numbers are preliminary.
- **One annotator for validity.** The validity check used one LLM judge, on 953 pairs from 8 runs in three strata, so its shares describe the sample and not the population of pairs. The judge is self-consistent ($\kappa = 0.92$), but self-consistency is not validity. There is no human annotator and no second model. The non-LLM signals (subject headings, citations) are partial, and the citation signal is thin for recent preprints.
- **Science-heavy corpus.** 86.4% of the corpus is computer science, mathematics or physics. Humanities, law, arts and general nonfiction appear only in the 500-record holdout, which is itself 76.4% class P, and

in the sample of 3,000 books, whose graph text is mostly LC subject headings. Domain schemes (arXiv, MSC, ACM CCS) are scored on their home ground and general schemes are not, so no cross-type comparison is fair.

- **Embedding as the only similarity.** The graph is always built from a text embedding. The sweep covers three models and three values of k, and a similarity from citations, co-use or subject headings could reorder results. The fitted scheme of Section 10 is the most exposed to this circularity.
- **Low gold coverage for MSC and JEL.** Only 13.2% of papers carry a valid MSC code and 0.02% a JEL code. Unplaced items hang under the root, so these gold scores measure coverage, not structure. The covered-subset analysis (Section 8.1) is the control, and its subset scores use a restricted graph.
- **ACM CCS gold missing, and no official crosswalk.** The corpus carries CCS 1998 codes, the scheme is CCS 2012, and ACM publishes no 1998-to-2012 mapping: it says it built one for its digital library and exposes it only as terms on each article's page, and the published CCS 2012 SKOS file holds no mapping or 1998 codes (Section 2). The gold run was skipped, and we did not build a mapping, because any we made would be ours and not ACM's.
- **Scheme licensing and the DDC position.** Several schemes cannot be redistributed: ACM CCS (educational and research use only), JEL (no open licence), MSC (non-commercial, share-alike), EuroVoc (terms stated inconsistently) and ILC2 (conflicting licence statements) (Table 1). Their files are built locally from the publishers' downloads and verified by hash, but are not published. Exact reproduction therefore depends on the upstream downloads staying available and unchanged. DDC is owned by OCLC [17]. We use Dewey class *numbers* from public catalogue records, and the tree is the notation's digit prefixes. No Dewey caption, schedule, table, Manual entry or relative-index term from any edition is used, and nothing is taken from OCLC services. The position rests on class numbers being short numeric strings in public catalogue data. It is an engineering position, not legal advice, and it is written for the United States and the European Union only. The Dewey-derived scheme that can be bundled and used to *place* items is FDC [20]. Even LCC's open status depends on jurisdiction (Section 2).
- **Titles-only holdout.** The book holdout has no descriptions. Its similarity graph is built from titles averaging 3.5 words, and the classifier finds candidates for few of them. The holdout is also small ($n = 500$). The main tables give point values; intervals for the placed runs are in the bootstrap runs of the covered and semantic result directories.
- **Navigation mixes coverage with structure.** Unplaced items sit under the root, so low coverage inflates root fan-out and every navigation cost. Navigation columns must be read next to coverage (Table 6).
- **Method limits.** The kNN-sparsified cost agrees with a dense cost on random subsets (Section 8.1) but has not been checked on the full corpus. Polyhierarchies are reduced to one parent per class. Only one placement per item is used, although 35.5% of a 200-work Open Library sample of English documents had more than one LCC value. The balance statistics count each item at every ancestor class (Section 3.5). The navigation models are unfitted. The sample of 3,000 books is stratified equally over LCC's main classes, which favours LCC's false-join term.
- **Self-embedding circularity of the fitted scheme.** Its insertion and the graph share an embedding, its corpus rows are in sample, and its labels were written by the AI coding assistant that built it and have not been evaluated by people (Section 10).
- **Search for prior work.** The novelty claim rests on the search in project report 8: 94 Crossref queries over the JASIS/JASIST family, Knowledge Organization, Cataloging & Classification Quarterly, ISKO, SIG/CR and NASKO proceedings, JCDL and TPDL/ECDL, plus five web searches. It screened titles and abstracts only. Full texts and non-English literature were not searched, one person screened, and Crossref's coverage of proceedings is incomplete (ISKO proceedings before 2012 are missing).

13 REPRODUCIBILITY

All inputs and outputs are in the project repository or pinned by hash.

- **Corpus.** `bench/corpus/corpus.jsonl` and `holdout.jsonl`, with ID lists and SHA-256 hashes. Metadata only, CC0 1.0. The original runs used a `corpus.jsonl` with SHA-256 prefix `7685685d`; the corrected label fields used for the MSC gold rescoring give prefix `1027bf9d`. Neither change touches row order, identifier, title or abstract, the only inputs to the graph.
- **Similarity graphs.** Published gzipped, with manifests, under `bench/results/2026-10-10/graphs/`. The manifests give the model revision `d13954661f83248295ba75c1ed411eef3b7b936e`, the `model.onnx` SHA-256, τ, k and the graph SHA-256 (corpus graph prefix `759d3b8461a9384d`).
- **Harness.** `crates/bench`, at the commit recorded in each `results.json` (`092494850adc`). The classifier for the placed run is at `f4ce24b78eda` (`profile:lexical@2`).
- **Schemes.** Bundled schemes are in the repository. Fetch-only schemes are rebuilt from the publishers' downloads, checked against committed `sources.lock.json` files, and must match the `file_sha256` recorded in `results.json`.
- **Commands.** Each result directory has a `REPRODUCE.md` listing every command. The first run took about 45 minutes and the rerun about 25 minutes, both on 8 CPUs.

- **Later results.** `docs/validity/` (sample, rubric, labels and tables) and `bench/results/2026-10-10-covered`, `-embed`, `-semantic`, `-learned`, `2026-10-11-falsejoin`, `-stability` and `-book-stability`, each with a `REPRODUCE.md`. Rerunning the judge needs the same model and rubric.
- **Determinism.** With the same graph and scheme files, gold scores reproduce exactly. Placed scores reproduce exactly with the same classifier build. Rebuilding the graph from text on a different ONNX Runtime version can change near-tied neighbours, so the published graph file, not the embedding pipeline, is the reproducible artefact.
- **This paper.** `paper/method-survey/scripts/gen_tables.py` regenerates every table and every headline value (the dictionary `v` in `generated.typ`) from the result files, with `paper/common/extra_results.py` for the validity, stability, book and semantic-tier values; `--check` fails if `generated.typ` is stale. Table 1 and the prose of Section 2 are not generated: they are taken from the scheme READMEs and the survey and licensing reports.

ACKNOWLEDGEMENTS

The author thanks Hetzner for offering an API for autonomous server provisioning; DeepSeek for releasing open-source models; and Anthropic for Claude, which assisted with the analysis, code and drafting.

REFERENCES

[1] M. Nocetti, "Agricultural soil science in universal classification systems. A comparative analysis," *Knowledge Organization*, vol. 5, no. 1, pp. 15–20, 1978, doi: 10.5771/0943-7444-1978-1-15.

[2] C. Zins and P. L. V. A. C. Santos, "Mapping the knowledge covered by library classification systems," *Journal of the American Society for Information Science and Technology*, vol. 62, no. 5, pp. 877–901, 2011, doi: 10.1002/asi.21481.

[3] D.-G. Oh, "Comparative Analysis of National Classification Systems: Cases of Korean Decimal Classification (KDC) and Nippon Decimal Classification (NDC)," *Knowledge Organization*, vol. 50, no. 2, pp. 83–98, 2023, doi: 10.5771/0943-7444-2023-2-83.

[4] Y. Wu, "Organization of Complex Topics in Comprehensive Classification Schemes: Case Studies of Disaster and Security," *Knowledge Organization*, vol. 50, no. 2, pp. 99–110, 2023, doi: 10.5771/0943-7444-2023-2-99.

[5] C. W. Cleverdon, "The ASLIB Cranfield research project on the comparative efficiency of indexing systems," *Aslib Proceedings*, vol. 12, no. 12, pp. 421–431, 1960, doi: 10.1108/eb049778.

[6] E. M. Keen, "The Aberystwyth index languages test," *Journal of Documentation*, vol. 29, no. 1, pp. 1–35, 1973, doi: 10.1108/eb026547.

[7] A. M. Carpenter, M. Jones, and C. Oppenheim, "Retrieval tests on five classification schemes," *Knowledge Organization*, vol. 5, no. 2, pp. 73–80, 1978, doi: 10.5771/0943-7444-1978-2-73.

[8] E. T. O'Neill, M. Dillon, and D. Vizine-Goetz, "Class dispersion between the Library of Congress Classification and the Dewey Decimal Classification," *Journal of the American Society for Information Science*, vol. 38, no. 3, pp. 197–205, 1987, doi: 10.1002/(sici)1097-4571(198705)38:3<197::aid-asi8>3.0.co;2-2.

[9] L. S. Connaway and M. C. Sievert, "Comparison of Three Classification Systems for Information on Health Insurance," *Cataloging & Classification Quarterly*, vol. 23, no. 2, pp. 89–104, 1996, doi: 10.1300/j104v23n02_07.

[10] OCLC, "Why did OCLC discontinue OCLC Classify?." [Online]. Available: https://help.oclc.org/Librarian_Toolbox/Troubleshooting/Why_is_OCLC_discontinuing_OCLC_Classify

[11] National Library of Finland, "Annif: automated subject indexing toolkit." [Online]. Available: https://github.com/NatLibFi/Annif

[12] K. Golub, O. Suominen, A. T. Mohammed, H. Aagaard, and O. Osterman, "Automated Dewey Decimal Classification of Swedish library metadata using Annif software," *Journal of Documentation*, vol. 80, no. 5, pp. 1057–1079, 2024, doi: 10.1108/JD-01-2022-0026.

[13] X. Song, P. Luo, J. Thomale, O. Zavalina, and L. Hong, "Comparative analysis of large language models' performance in book classification tasks using Library of Congress Classification system," *Journal of Information Science*, 2026, doi: 10.1177/01655515261425547.

[14] C. Holstrom, "Large Language Models (LLMs) and Cataloging: Exploring How ChatGPT and Copilot Assign Subject Headings and Call Numbers," *NASKO*, vol. 10, pp. 78–95, 2025, doi: 10.7152/nasko.v7i1.95648.

[15] G. K. Shahi, R. Shigapov, and O. Hummel, "LLM4DDC: Adopting Large Language Models for Research Data Classification Using Dewey Decimal Classification," in *E-Science-Tage 2025: Research Data Management: Challenges in a Changing World*, V. Heuveline, P. Kling, F. Heuschkel, S. G. Habinger, and C. F. Krömer, Eds., Heidelberg: heiBOOKS, 2025, pp. 476–484. doi: 10.11588/heibooks.1652.c23948.

[16] C. Saccucci and A. Potter, "Results of AI Experimentation for Cataloging at the Library of Congress." [Online]. Available: https://repository.ifla.org/handle/20.500.14598/6861

[17] OCLC, "Dewey Services – Resources." [Online]. Available: https://www.oclc.org/en/dewey/resources.html

[18] UDC Consortium, "UDC licences and terms." [Online]. Available: https://udcc.org/licences_terms.htm

[19] BARTOC, "Basisklassifikation (Basic Classification, BK)." [Online]. Available: https://bartoc.org/en/node/18785

[20] J. M. Ockerbloom, "Free Decimal Correspondence." [Online]. Available: https://everybodyslibraries.com/free-decimal-correspondence/

[21] ISKO Italia, "Integrative Levels Classification, edition 2." [Online]. Available: https://www.iskoi.org/ilc/

[22] OpenAlex, "Topics." [Online]. Available: https://help.openalex.org/hc/en-us/articles/24736129405719-Topics

[23] G. Jahoda, "ASLIB Cranfield Research Project, Report on the First Stage of an Investigation into the Comparative Efficiency of Indexing Systems (Book Review)," *College & Research Libraries*, vol. 22, no. 3, p. 228, 1961, doi: 10.5860/crl_22_03_228.

[24] I. Rafols and L. Leydesdorff, "Content-based and algorithmic classifications of journals: Perspectives on the dynamics of scientific communication and indexer effects," *Journal of the American Society for Information Science and Technology*, vol. 60, no. 9, pp. 1823–1835, 2009, doi: 10.1002/asi.21086.

[25] Q. Wang and L. Waltman, "Large-scale analysis of the accuracy of the journal classification systems of Web of Science and Scopus," *Journal of Informetrics*, vol. 10, no. 2, pp. 347–364, 2016, doi: 10.1016/j.joi.2016.02.003.

[26] R. Klavans and K. W. Boyack, "Which Type of Citation Analysis Generates the Most Accurate Taxonomy of Scientific and Technical Knowledge?," *Journal of the Association for Information Science and Technology*, vol. 68, no. 4, pp. 984–998, 2017, doi: 10.1002/asi.23734.

[27] C.-A. Julien, P. Tirilly, J. D. Dinneen, and C. Guastavino, "Reducing subject tree browsing complexity," *Journal of the American Society for Information Science and Technology*, vol. 64, no. 11, pp. 2201–2223, 2013, doi: 10.1002/asi.22915.

[28] C. Mader, B. Haslhofer, and A. Isaac, "Finding Quality Issues in SKOS Vocabularies," in *Theory and Practice of Digital Libraries*, in Lecture Notes in Computer Science., Springer Berlin Heidelberg, 2012, pp. 222–233. doi: 10.1007/978-3-642-33290-6_25.

[29] L. A. Owens and P. A. Cochrane, "Thesaurus Evaluation," *Cataloging & Classification Quarterly*, vol. 37, no. 3–4, pp. 87–102, 2004, doi: 10.1300/j104v37n03_07.

[30] S. Dasgupta, "A cost function for similarity-based hierarchical clustering," in *Proceedings of the 48th Annual ACM Symposium on Theory of Computing (STOC)*, 2016, pp. 118–127. doi: 10.1145/2897518.2897527.

[31] B. Moseley and J. R. Wang, "Approximation Bounds for Hierarchical Clustering: Average Linkage, Bisecting K-means, and Local Search," in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. [Online]. Available: https://proceedings.neurips.cc/paper/2017/hash/d8d31bd778da8bdd536187c36e48892b-Abstract.html

[32] B. Moseley and J. R. Wang, "Approximation Bounds for Hierarchical Clustering: Average Linkage, Bisecting K-means, and Local Search," *Journal of Machine Learning Research*, vol. 24, no. 1, pp. 1–36, 2023, [Online]. Available: https://jmlr.org/papers/v24/18-080.html

[33] M. Charikar and V. Chatziafratis, "Approximate Hierarchical Clustering via Sparsest Cut and Spreading Metrics," in *Proceedings of the Twenty-Eighth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2017, pp. 841–854. doi: 10.1137/1.9781611974782.53.

[34] A. Roy and S. Pokutta, "Hierarchical Clustering via Spreading Metrics," in *Advances in Neural Information Processing Systems 29 (NeurIPS 2016)*, 2016. [Online]. Available: https://papers.nips.cc/paper_files/paper/2016/hash/4d2e7bd33c475784381a64e43e50922f-Abstract.html

[35] V. Cohen-Addad, V. Kanade, F. Mallmann-Trenn, and C. Mathieu, "Hierarchical Clustering: Objective Functions and Algorithms," *Journal of the ACM*, vol. 66, no. 4, pp. 1–42, 2019, doi: 10.1145/3321386.

[36] I. Chami, A. Gu, V. Chatziafratis, and C. Ré, "From Trees to Continuous Embeddings and Back: Hyperbolic Hierarchical Clustering," in *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020, pp. 15065–15076. [Online]. Available: https://papers.nips.cc/paper_files/paper/2020/hash/ac10ec1ace51b2d973cd87973a98d3ab-Abstract.html

[37] V. Chatziafratis, R. Niazadeh, and M. Charikar, "Hierarchical Clustering with Structural Constraints," in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, in Proceedings of Machine Learning Research, vol. 80. 2018, pp. 774–783. [Online]. Available: https://proceedings.mlr.press/v80/chatziafratis18a.html

[38] scikit-network developers, "scikit-network: hierarchy module (dasgupta_cost, dasgupta_score)." [Online]. Available: https://scikit-network.readthedocs.io/en/latest/reference/hierarchy.html

[39] J. M. Kleinberg, "An Impossibility Theorem for Clustering," in *Advances in Neural Information Processing Systems 15 (NIPS 2002)*, 2002. [Online]. Available: https://papers.nips.cc/paper_files/paper/2002/hash/43e4e6a6f341e00671e123714de019a8-Abstract.html

[40] G. Carlsson and F. Mémoli, "Characterization, Stability and Convergence of Hierarchical Clustering Methods," *Journal of Machine Learning Research*, vol. 11, no. 47, pp. 1425–1470, 2010, [Online]. Available: https://www.jmlr.org/papers/v11/carlsson10a.html

[41] D. A. Huffman, "A Method for the Construction of Minimum-Redundancy Codes," *Proceedings of the IRE*, vol. 40, no. 9, pp. 1098–1101, 1952, doi: 10.1109/JRPROC.1952.273898.

[42] D. E. Knuth, "Optimum binary search trees," *Acta Informatica*, vol. 1, no. 1, pp. 14–25, 1971, doi: 10.1007/BF00264289.

[43] D. J. Watts, P. S. Dodds, and M. E. J. Newman, "Identity and Search in Social Networks," *Science*, vol. 296, no. 5571, pp. 1302–1305, 2002, doi: 10.1126/science.1070120.

[44] E. Rosch, C. B. Mervis, W. D. Gray, D. M. Johnson, and P. Boyes-Braem, "Basic objects in natural categories," *Cognitive Psychology*, vol. 8, no. 3, pp. 382–439, 1976, doi: 10.1016/0010-0285(76)90013-X.

[45] K. Snowberry, S. R. Parkinson, and N. Sisson, "Computer display menus," *Ergonomics*, vol. 26, no. 7, pp. 699–712, 1983, doi: 10.1080/00140138308963390.

[46] P. G. Zaphiris, "Depth vs Breath in the Arrangement of Web Links," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 44, no. 4, pp. 453–456, 2000, doi: 10.1177/154193120004400414.

[47] D. P. Miller, "The Depth/Breadth Tradeoff in Hierarchical Computer Menus," *Proceedings of the Human Factors Society Annual Meeting*, vol. 25, no. 1, pp. 296–300, 1981, doi: 10.1177/107118138102500179.

[48] K. Larson and M. Czerwinski, "Web page design," in *Proceedings of the SIGCHI conference on Human factors in computing systems - CHI '98*, ACM Press, 1998, pp. 25–32. doi: 10.1145/274644.274649.

[49] T. K. Landauer and D. W. Nachbar, "Selection from alphabetic and numeric menu trees using a touch screen," in *Proceedings of the SIGCHI conference on Human factors in computing systems - CHI '85*, ACM Press, 1985, pp. 73–78. doi: 10.1145/317456.317470.

[50] B.-A. Lipetz and C. T. Song, "How many cards per file guide? optimizing the two-level file," *Journal of the American Society for Information Science*, vol. 21, no. 2, pp. 140–141, 1970, doi: 10.1002/asi.4630210206.

[51] L. Waltman and N. J. van Eck, "A new methodology for constructing a publication-level classification system of science," *Journal of the American Society for Information Science and Technology*, vol. 63, no. 12, pp. 2378–2392, 2012, doi: 10.1002/asi.22748.

[52] V. A. Traag, L. Waltman, and N. J. van Eck, "From Louvain to Leiden: guaranteeing well-connected communities," *Scientific Reports*, vol. 9, p. 5233, 2019, doi: 10.1038/s41598-019-41695-z.

[53] J. Priem, H. Piwowar, and R. Orr, "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts." [Online]. Available: https://arxiv.org/abs/2205.01833

[54] C. Zhang *et al.*, "TaxoGen: Unsupervised Topic Taxonomy Construction by Adaptive Term Embedding and Clustering," in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*, 2018, pp. 2701–2709. doi: 10.1145/3219819.3220064.

[55] J. Shen, Z. Shen, C. Xiong, C. Wang, K. Wang, and J. Han, "TaxoExpan: Self-supervised Taxonomy Expansion with Position-Enhanced Graph Neural Network," in *Proceedings of The Web Conference 2020*, 2020, pp. 486–497. doi: 10.1145/3366423.3380132.

[56] Q. Zeng *et al.*, "Chain-of-Layer: Iteratively Prompting Large Language Models for Taxonomy Induction from Limited Examples," in *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM)*, 2024, pp. 3093–3102. doi: 10.1145/3627673.3679608.

[57] M. Wan *et al.*, "TnT-LLM: Text Mining at Scale with Large Language Models," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2024, pp. 5836–5847. doi: 10.1145/3637528.3671647.

[58] P. Kargupta, N. Zhang, Y. Zhang, R. Zhang, P. Mitra, and J. Han, "TaxoAdapt: Aligning LLM-Based Multidimensional Taxonomy Construction to Evolving Research Corpora," in *Proceedings of the 63rd Annual Meeting of the Association for*

Computational Linguistics (Volume 1: Long Papers), 2025, pp. 29834–29850. doi: 10.18653/v1/2025.acl-long.1442.

[59] M. Unterkalmsteiner and W. Abdeen, "A compendium and evaluation of taxonomy quality attributes," *Expert Systems*, vol. 40, no. 1, 2023, doi: 10.1111/exsy.13098.

[60] P. Wullschleger, M. Zarharan, D. Daly, M. Pouly, and J. Foster, "Reference-Free Evaluation of Taxonomies," in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 25489–25507. doi: 10.18653/v1/2026.findings-acl.1273.

[61] O. Suominen, J. Inkinen, and M. Lehtinen, "Annif at SemEval-2025 Task 5: Traditional XMTC augmented by LLMs," in *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*, 2025, pp. 2424–2431. [Online]. Available: https://arxiv.org/abs/2504.19675

[62] J. D'Souza, S. Sadruddin, H. Israel, M. Begoin, and D. Slawig, "SemEval-2025 Task 5: LLMs4Subjects – LLM-based Automated Subject Tagging for a National Technical Library's Open-Access Catalog." [Online]. Available: https://arxiv.org/abs/2504.07199

[63] ahmad-PH, "auto_lcc: Automatic Library of Congress Classification." [Online]. Available: https://github.com/ahmad-PH/auto_lcc

[64] Library of Congress, "Library of Congress Classification PDF Files." [Online]. Available: https://www.loc.gov/aba/publications/FreeLCC/freelcc.html

[65] Library of Congress, "About: LC Linked Data Service." [Online]. Available: https://id.loc.gov/about/

[66] Legal Information Institute, "17 U.S. Code § 105 – Subject matter of copyright: United States Government works." [Online]. Available: https://www.law.cornell.edu/uscode/text/17/105

[67] Library of Congress, "Classification Web: Legal Notices." [Online]. Available: https://classweb.loc.gov/legal.html

[68] C. Fraser, "Mathematics in Library and Review Classification Systems: An Historical Overview," *Knowledge Organization*, vol. 47, no. 4, pp. 334–356, 2020, doi: 10.5771/0943-7444-2020-4-334.

[69] OpenAlex, "Pricing Overview." [Online]. Available: https://help.openalex.org/hc/en-us/articles/24397762024087-Pricing

[70] WIPO, "International Patent Classification, edition 2026.01, master files." [Online]. Available: https://www.wipo.int/classifications/data/ipc/ITSupport_and_download_area/20260101/MasterFiles/ipc_scheme_20260101.zip

[71] WIPO, "Terms of Use: Open access policy." [Online]. Available: https://www.wipo.int/en/web/terms-of-use

[72] arXiv, "arXiv Category Taxonomy." [Online]. Available: https://arxiv.org/category_taxonomy

[73] Mathematical Reviews and zbMATH Open, "Mathematics Subject Classification MSC2020." [Online]. Available: https://msc2020.org/

[74] American Economic Association, "JEL Classification System." [Online]. Available: https://www.aeaweb.org/econlit/jelCodes.php

[75] ACM, "The 2012 ACM Computing Classification System." [Online]. Available: https://web.archive.org/web/20241222222257/https://www.acm.org/publications/class-2012

[76] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3980–3990. doi: 10.18653/v1/D19-1410.

[77] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers." [Online]. Available: https://arxiv.org/abs/2002.10957

[78] sentence-transformers, "all-MiniLM-L6-v2." [Online]. Available: https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[79] Qdrant, "all-MiniLM-L6-v2 (ONNX export), revision d13954661f83248295ba75c1ed411eef3b7b936e." [Online]. Available: https://huggingface.co/Qdrant/all-MiniLM-L6-v2-onnx

[80] arXiv, "arXiv License Information." [Online]. Available: https://info.arxiv.org/help/license/index.html

[81] Open Library, "Developer Center / Licensing." [Online]. Available: https://openlibrary.org/developers/licensing

[82] Open Library, "Using Open Library Data." [Online]. Available: https://openlibrary.org/help/faq/using

[83] J. Cohen, "A Coefficient of Agreement for Nominal Scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960, doi: 10.1177/001316446002000104.

[84] B. Efron, "Bootstrap Methods: Another Look at the Jackknife," *The Annals of Statistics*, vol. 7, no. 1, 1979, doi: 10.1214/aos/1176344552.

[85] T. C. Hu and A. C. Tucker, "Optimal Computer Search Trees and Variable-Length Alphabetical Codes," *SIAM Journal on Applied Mathematics*, vol. 21, no. 4, pp. 514–532, 1971, doi: 10.1137/0121057.